

April 2006



Product Comparison

Ventilators, Intensive Care

UMDNS information

This Product Comparison covers the following device term and product code as listed in ECRI's Universal Medical Device Nomenclature System™ (UMDNS™):

- ✓ Ventilators, Intensive Care [17-429]
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Ventilators, Intensive Care

Scope of this Product Comparison

This Product Comparison covers positive-pressure ventilators designed for use in critical care settings. Although some of the models listed in the chart may have high-frequency ventilation capabilities, units that produce only high-frequency ventilation are not covered, nor are dedicated neonatal/pediatric ventilators, transport ventilators, anesthesia ventilators, or ventilators designed exclusively for home care. For information on these types of ventilators, see the following Product Comparisons:

- Anesthesia Units
- Ventilators, Intensive Care, Neonatal/Pediatric
- Ventilators, Portable
- Ventilators, Transport

These devices are also called: critical care ventilators, continuous ventilators, intensive care ventilators, positive-pressure ventilators.

Purpose

Ventilators provide temporary ventilatory support or respiratory assistance to patients who cannot breathe on their own or who require assistance to maintain adequate ventilation because of illness, trauma, congenital defects, or drugs (e.g., anesthetics). In most mechanical ventilators, a positive-pressure source delivers gas to the patient's lungs to support gas exchange; to open or maintain ventilation of alveoli, where gas exchange occurs; and to rest ventilatory muscles until the patient is able to safely resume adequate spontaneous ventilation. Positive-pressure breaths are typically delivered through an endotracheal tube or a tracheostomy tube. The pressure in the lungs increases in proportion to the volume of inflating gas. The pressure is relieved as gas is exhaled through an exhalation pathway.

A high-frequency ventilator uses positive pressure to deliver breaths at frequencies much higher than the normal breathing rate (e.g., >100 breaths/min). High-frequency ventilators were developed in an effort to reduce barotrauma and some of the deleterious hemodynamic effects associated with the high tidal volumes and positive pressure used with conventional ventilators. These ventilators are available for patients who cannot tolerate the airway pressures needed for ventilation at typical volumes; they may also be used when a motionless field is required for surgery near the airways.



Principles of operation

A critical care ventilator (see Figure 1) typically consists of a flexible breathing circuit, a control system, a gas supply, and monitors and alarms. Heating and humidification devices are available as add-on components. Most ventilators are microprocessor controlled and regulate the pressure, volume, or flow of the delivered positive-pressure breath, as well as the fraction of inspired oxygen (FiO_2), based on control settings. Communications interfaces are also typically included so that information on control settings, monitored variables, and alarm status can be transferred to a bedside monitor, an information system, or some other interfaced device. Power is supplied from either an electrical wall outlet or a battery; battery power is used for short-term ventilation, such as during intrahospital patient transport.

Some intensive care ventilators can receive gas (both air and oxygen [O_2]) from a wall outlet that generally provides gas at a pressure of approximately 50 pounds per square inch (psi); the flow of gas to the patient can be regulated by flow-control valves on the ventilator. To obtain the desired FiO_2 for delivery to the patient, most ventilators mix air and O_2 internally. During inspiratory gas delivery, an exhalation valve is closed to maintain pressure in the breathing circuit and lungs. During expiratory gas delivery, an exhalation valve is opened to release gas to ambient air.

The gas is delivered to the patient through the flexible breathing circuit.

Most intensive care ventilators use a double-limb breathing circuit made of corrugated plastic tubing to transport the gas from the ventilator to the patient and return the exhaled gas to the ventilator through one of the limbs (referred to as the expiratory limb). During inspiratory gas delivery, the exhalation valve is closed to maintain pressure in the breathing circuit and lungs. After the inspiratory phase, the gas is released to ambient air through this valve. The breathing circuit also provides sites where the delivered gas may be heated; humidified; monitored for proximal airway pressure; and conditioned with nebulized medications, as well as where condensation may be collected. Many models have sensors within the ventilator or breathing circuit that can measure airway pressure or flow and provide feedback to the ventilator to automatically adjust its output.

Controls

Controls are used to select breathing mode and ventilation pattern parameters (e.g., tidal volume, breathing rate). For the ventilator to produce a prescribed breathing pattern, several parameters can be independently set, such as length of the inspiratory or expiratory phase, rate of mechanical breaths, ratio of inspiratory time to expiratory time (I:E ratio), waveform shape, tidal volume, minute volume (the volume inhaled in one minute), peak inspiratory flow, peak pressure, and positive end-expiratory pressure (PEEP).

Intensive care ventilators have continuous positive airway pressure (CPAP) and PEEP controls, which are regulated by a restriction of flow to the exhalation valve. CPAP provides continuous positive airway pressure in the breathing circuit as the patient breathes spontaneously. This keeps the alveoli and airways inflated by preventing proximal airway pressure from returning to zero at the end of exhalation. CPAP is applied to patients who can breathe spontaneously and do not require full ventilatory support. It can improve lung volume and, consequently, oxygenation and lung function by increasing alveolar volumes, recruitment, and stability. By helping to redistribute interstitial water, CPAP also improves O_2 diffusion across the alveolar capillary membrane. CPAP

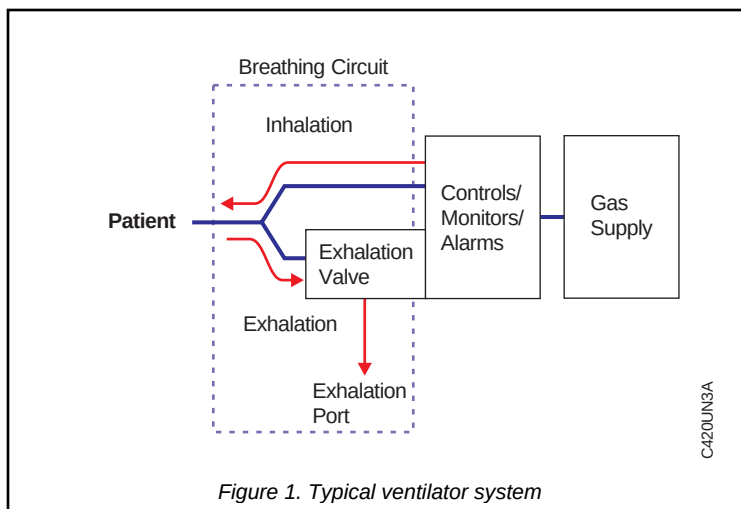


Figure 1. Typical ventilator system

may be used to raise the patient's arterial partial pressure of oxygen (PaO_2) without requiring an increase in the FiO_2 . PEEP maintains a positive airway pressure from the end of an assisted, controlled, spontaneous, or mandatory exhalation to the beginning of the next inspiration. The result is similar to that obtained with CPAP and is achieved by restricting or prohibiting the exhalation of gases through the exhalation valve after the pressure has dropped to a prescribed value. PEEP aids in maintaining lung volume and in preventing alveolar collapse. Increases in PEEP are often used to increase the patient's arterial O_2 saturation without increasing the inspired O_2 percentage, although very high PEEP may decrease venous return, cardiac output, and O_2 transport and increase pulmonary vascular resistance.

The I:E ratio is an indication of the partitioning of a breath into inspiration and expiration. In general, the expiratory time is set to be longer than the inspiratory time (e.g., I:E ratio is 1:2); however, an inverse ratio can also be used so that the inspiratory time exceeds the expiratory time (e.g., I:E ratio is 1:0.5). Because inverse I:E settings are not normally used, some ventilators signal when an inverse I:E ratio has been reached; others will not deliver inverse I:E breaths.

Controls are also available for setting the flow waveform. Volume-controlled ventilation flow generally has square, accelerating, decelerating, or sinusoidal waveforms. Pressure ramp adjustments are now available in pressure-controlled ventilation modes. Such adjustments allow the user to maximize the flow and pressure levels while maintaining a flow delivery that lowers the work of breathing and is more comfortable for the patient.

Operating modes

Intensive care ventilators have several operating modes; a mode of operation defines the algorithm that will be used to initiate (or trigger) and end a machine breath. Different modes can provide either full or partial ventilatory support, depending on the individual patient's condition and respiratory ability.

The control mode provides full support to patients who cannot breathe for themselves; it is infrequently used. In this mode, the ventilator provides mandatory breaths at preset time intervals and does not allow the patient to breathe spontaneously. This mode requires the patient to be unconscious or sedated to stop spontaneous attempts to breathe asynchronously with the ventilator.

Assist/control modes also provide full support by delivering an assisted breath whenever the ventilator senses a patient's inspiratory effort and by delivering mandatory breaths at preset time intervals. This mode is designed for patients who have difficulty breathing but can initiate inspiration. A breath is triggered when a patient's breathing efforts are detected as a drop in pressure in the breathing circuit (pressure triggering) or as a difference in flow between the circuit's inspiratory and expiratory limbs (flow triggering). Ventilators can use one or both of these methods of triggering.

Most intensive care ventilators can deliver volume- and pressure-controlled breaths that can be used to provide both full and partial ventilatory support. Volume-controlled breaths of ventilation use a control system to ensure that a set tidal volume is delivered during the inspiratory cycle. The set volume may not be delivered if the pressure exceeds the setting of the high-pressure alarm or of a pressure-relief valve. These modes are typically used for adult and pediatric patients to maintain



adequate pulmonary gas exchange. Pressure-controlled breaths regulate flow delivery to attain and sustain a clinician-set inspiratory pressure level for a set time so that the ventilator delivers controlled or assisted breaths that are time cycled. Combination modes, which are now available on several models, combine volume- and pressure-controlled ventilation to ensure that a minimum volume is delivered with an initial flow that matches patient demand. These modes may allow more effective ventilation of patients whose lung characteristics change frequently.

The synchronized intermittent mandatory ventilation (SIMV) mode delivers controlled breaths at a set frequency and also allows the patient to breathe spontaneously with no assistance during the periods between the controlled breaths. The mandatory breaths in this mode are synchronized with a spontaneous breathing effort if that effort occurs sufficiently close to the time the mandatory breath would have been produced. This reduces the possibility of overinflation, which could result from stacking a mandatory breath on a spontaneous breath.

Pressure support reduces the work of spontaneous breathing by delivering a preset level of positive pressure to the patient's airway during a spontaneous inspiratory effort. This reduces the work of the patient's respiratory muscles and minimizes the effort needed to draw an adequate amount of air into the lungs. It also compensates for the extra work of breathing imposed by the ventilator tubing and valves. Pressure support is an adjunct that may be added to spontaneous efforts in the SIMV and CPAP modes.

Another mode of ventilation is airway pressure release ventilation (APRV). APRV may be used to treat acute lung injury in patients who require mechanical support. Once an appropriate level of CPAP is selected, APRV is initiated whenever mechanical assistance is required by cyclically releasing the airway pressure until a lower level is obtained. Carbon dioxide exits the lungs passively as the airway pressure decreases. When this brief release period ends, the airway pressure rapidly returns to the CPAP level. This mode provides mechanical ventilatory assistance without raising the airway pressure above the CPAP level; consequently, barotrauma and adverse hemodynamic effects may occur less frequently than with other conventional modes of mechanical ventilation.

Monitors and alarms

Intensive care ventilators are equipped with a variety of monitors and alarms to detect equipment-related problems and changes in patient status, to ensure that the user adjusts settings to achieve effective ventilation, and to reduce the risk of ventilator-induced injury (e.g., barotrauma). Variables that are typically monitored and displayed on the ventilator include the following:

- A continuous indication of airway pressure, as well as peak, mean, and baseline pressures
- Mechanical and spontaneous respiratory rates
- I:E ratio
- O₂ concentration
- Exhaled volumes of mechanical and spontaneous breaths (tidal volumes) and accumulated volume over one minute (exhaled minute volume)

Graphics monitors include graphs of pressure, volume, and flow versus time. To track the patient's progress, the monitor calculates patient pulmonary mechanics (e.g., compliance, resistance) from monitored variables. Pressure-volume loops, which are graphs of pressure versus volume over a single breath, and flow-volume loops, which are graphs of flow versus volume over a single breath, indicate breathing abnormalities such as obstructive or restrictive lung disease. Graphics monitors are generally part of the ventilator unit. If not, the manufacturer usually sells one with the unit.

Graphics monitors allow the clinician to optimize ventilator settings values and assist with diagnostics. The displayed loops and lung mechanics parameters can also help clinicians recognize obstructive or restrictive flow patterns.

Because the consequences of incorrect or inadequate mechanical ventilation can be severe, ventilators are equipped with audible and visual alarms to notify clinicians of changes in the patient's condition or of device problems. Most ventilators have alarms for apnea, high and low

respiratory rate, high and low pressure, loss of power, loss of high-pressure gas, system malfunctions, incorrect O₂ concentration, and exhaled volume. Some ventilators also have baseline-pressure alarms; the low-baseline-pressure alarm alerts clinicians to losses of PEEP, which can affect the patient's O₂ saturation, and the high-baseline-pressure alarm alerts clinicians to inadvertent increases in PEEP, which can prevent complete exhalation. An alarm should also be activated if disconnections occur in the breathing circuit or if flow resistance is encountered. A loss of power or the gas supply, or other conditions affecting a ventilator's ability to operate, should produce an alarm and allow the patient to spontaneously breathe air or the specified gas mixture. All critical alarms should be easy to identify and impossible to disarm indefinitely.

Additionally, to prevent injury to the patient until clinicians can respond to alarms, ventilators incorporate a number of safety features such as the ability to release pressure at the level of the high-pressure alarm setting. Another feature is the presence of backup ventilation, in which the ventilator will initiate breaths when it senses that the patient's breathing efforts have ceased.

Alarm-enhancement systems

Ventilator alarms are crucial for safeguarding the health and lives of patients. Therefore, it is vital that they be readily detected in even the busiest, noisiest hospital departments. Alarm-enhancement systems, which communicate ventilator alarms to locations where they are more likely to be detected by caregivers, can be helpful.

There are four basic categories of ventilator alarm enhancements:

- Interfacing ventilators with physiologic monitors
- Incorporating commercially available systems for centralized monitoring of ventilator alarms
- Interfacing ventilators with nurse call systems
- Utilizing remote annunciators for ventilator alarms

The various alarm-enhancement options range widely in complexity, cost, and the types of care settings for which they're likely to be suitable. For more information on alarm-enhancements systems and ventilator-physiologic monitoring system interfaces, see ECRI's Health Devices citation in this report.

Communication interfaces

Most intensive care ventilators have a standard or optional interface through which the ventilator can be connected to a bedside monitor or information system. Ventilator settings, monitored variables, and information on alarms can be transmitted through this interface. On some units, the interface can connect two ventilators, synchronizing them so that they can independently ventilate both lungs (e.g., for a patient with unilateral lung disease).

Reported problems

The most common problem associated with intensive care ventilators is the risk of a patient acquiring ventilator-associated pneumonia (VAP). It is generally accepted that prolonged ventilation periods greatly increase a patient's risk of acquiring VAP. The link between prolonged ventilation and VAP is unclear, but following proper infection control procedures in maintaining the ventilator, the breathing circuit, and all associated equipment can minimize patient risk.

Leaks in the breathing circuit or components may prevent the ventilator from delivering a preset tidal volume or accurately sensing flow and terminating a pressure-supported breath. Also, such leaks can affect the ventilator's ability to maintain the PEEP level. This in turn may affect O₂ saturation and can result in autocycling.

The friction-fit connector that attaches a ventilator to a patient's artificial airway can be accidentally disconnected if it is not attached securely by the clinician. Ventilators should signal an audible and visual alarm when they detect a leak or disconnection; however, some low-pressure alarms can be inappropriately adjusted below the detection threshold.

Patient-ventilator dyssynchrony refers to the situation in which a mechanically ventilated patient fails to trigger the ventilator, or the ventilator erroneously senses a patient's effort and delivers breaths. The result is a machine breath rate that is inappropriate to the rate of the patient's inspiratory efforts. This is also called trigger failure or desynchronization, mismatching, and "fighting the ventilator." One cause for patient-ventilator dyssynchrony is improper setting of trigger sensitivity. The use of airway pressure and flow waveforms to detect it may not be reliable, since airway pressure and flow measured at the ventilator can be affected by various artifacts (e.g., hiccups, coughs, sudden displacement or compression of the ventilator tubing). Clinical observation is highly specific in identifying patient-ventilator dyssynchrony, since observation of thoracoabdominal movement has been the standard method of determining respiratory rate, and patients with patient-ventilator dyssynchrony often have heightened and prominent accessory muscle activity associated with inspiratory efforts. When gas delivery is not synchronized with the patient's efforts to initiate a breath, increased patient discomfort and work of breathing can result. This can also lead to respiratory distress, can inhibit pulmonary gas exchange, and can make weaning the patient from mechanical ventilation more difficult.

Because in many cases the patient depends entirely on the ventilator for life support, ensuring proper maintenance and avoiding operator errors or machine failures can be critical. Some of the unavoidable risks of mechanical ventilation include barotrauma (when an airway pressure that is too high damages the lungs), reduced cardiac output, and adverse effects on gas exchange in the lungs. By using a well-designed ventilator and ensuring that it is set up and operated correctly, users can typically avoid injury to the patient. In addition, ventilators are typically inspected at least semiannually, and operation is usually verified (or should be) before each use. These procedures should detect most cases of inappropriate setup or mechanical problems.

Purchase considerations

ECRI recommendations

Included in the accompanying comparison chart are ECRI's recommendations for minimum performance requirements for intensive care ventilators. The requirements are separated into two categories—basic and mid/high complexity. The differences between these two categories are based on performance criteria for operating modes, controls, monitored parameters, and alarm functionality.

The ventilator should offer assist/control and SIMV modes. For volume and pressure breaths, it should also provide CPAP/PEEP and pressure support. The unit should monitor airway pressure, respiratory rate, I:E ratio, and minute volume; controls should be available for pressure level, tidal volume, breath rate, inspiratory time, FiO₂, PEEP/CPAP, I:E ratio, pressure support, and sensitivity.

ECRI recommends that these units have patient-responsive features and patient-responsive mode(s) or combination mode(s). For higher-end ventilators, their graphic displays should include waveforms and loops. Loops should be saved for comparisons, trending of monitored variables. The higher-end units should also offer some respiratory maneuvers (e.g., PO. 1).

Alarms, both visual and auditory, should be available for inspiratory pressure (low and high), low CPAP/PEEP, minute volume (low or low/high), respiratory rate (low and high), gas supply loss, and power failure. All alarms should be distinct and easily identified. Also, if alarm volume is adjustable, it should not be possible to turn the volume down so low that the alarm is inaudible. The alarm silence feature must reactivate automatically within two minutes if the condition is not corrected. If an alarm is silenced, a visual display should clearly indicate which alarm is disabled.

The delivered O₂ or O₂/air mixture should be monitored with an O₂ analyzer that includes an alarm for concentrations outside acceptable ranges. The analyzer should be included with the ventilator.

The controls (i.e., switches and knobs) should be visible and clearly identified, and their functions should be self-evident. The design should prevent misinterpretation of displays and control settings. Controls should be protected against accidental setting changes (e.g., due to someone brushing

against the panel) and be sealed against fluid penetration. Patient and operator safety and system performance should not be adversely affected by fluid spills.

Other considerations

Current ventilator designs offer an often complicated variety of options, requiring a knowledgeable user. Staff shortages and frequent employee turnover in some hospitals often make adequate formal training in the use of clinical equipment difficult. Therefore, ventilators with good human factors design offer a significant advantage. In addition, standardizing equipment helps minimize retraining and confusion, and suppliers often give significant discounts when multiple units are purchased.

A wide range of modes, variables monitored and controlled, and alarms is offered among different ventilators. These features should be evaluated to determine which are needed for a particular patient population and clinical setting.

Cost containment

Because intensive care ventilators entail ongoing maintenance and operational costs, the initial acquisition cost does not accurately reflect the total cost of ownership. Therefore, a purchase decision should be based on issues such as life-cycle cost (LCC), local service support, discount rates and non-price-related benefits offered by the supplier, and standardization with existing equipment in the department or hospital (i.e., purchasing all ventilators from one supplier).

An LCC analysis can be used to compare high-cost alternatives and/or to determine the positive or negative economic value of a single alternative. For example, hospitals can use LCC analysis techniques to examine the cost-effectiveness of leasing or renting equipment versus purchasing the equipment outright. Because it examines the cash-flow impact of initial acquisition costs and operating costs over a period of time, LCC analysis is most useful for comparing alternatives with different cash flows and for revealing the total costs of equipment ownership. One LCC technique—present value (PV) analysis—is especially useful because it accounts for inflation and for the time value of money (i.e., money received today is worth more than money received at a later date). Conducting a PV/LCC analysis often demonstrates that the cost of ownership includes more than just the initial acquisition cost and that a small increase in initial acquisition cost may produce significant savings in long-term operating costs. The PV is calculated using the annual cash outflow, the dollar discount factor (the cost of capital), and the lifetime of the equipment (in years) in a mathematical equation.

The following represents a sample seven-year PV/LCC analysis for an intensive care ventilator.

Present Value/Life-Cycle Cost Analysis

Assumptions

- Operating costs are considered for years 1 through 7
- Dollar discount factor is 6.5%
- Inflation rate is 4% for disposables, 6% for a full-service contract
- Disposable breathing circuits are changed once a week

Capital Costs

- Ventilator = \$27,000

Total Capital Costs = \$27,000

Operating Costs

- Disposables = \$1,040/year (\$20 disposable breathing circuits and accessories are changed once a week)
- Service contract, years 1 through 7 = \$1,100/year

Total Operating Costs = \$2,140/year

PV = (\$41,185)

Other costs not included in the above analysis that should be considered for budgetary planning include those associated with the following:

- Reusable breathing circuits
- Disinfection of reusable breathing circuits
- Parts replaced during preventive maintenance (e.g., sensors)
- Staff training
- Utilities
- Contributions to overhead

Clearly, the expected expense of operating an intensive care ventilator is significantly greater than the initial cost of the device. Hospitals should evaluate how they plan to use the ventilator; in particular, the decision to use disposable or reusable breathing circuits will affect the cost of operation.

Hospitals can purchase service contracts or service on a time-and-materials basis from the supplier. Service may also be available from a third-party organization. The decision to purchase a service contract should be carefully considered. Purchasing a service contract ensures that preventive maintenance will be performed at regular intervals, thereby eliminating the possibility of unexpected maintenance costs. Also, many suppliers do not extend system performance and uptime guarantees beyond the length of the warranty unless the system is covered by a service contract.

ECRI recommends that, to maximize bargaining leverage, hospitals negotiate pricing for service contracts before the system is purchased. Additional service contract discounts may be negotiable for multiple-year agreements or for service contracts that are bundled with contracts on other similar equipment in the department or hospital. For customized analyses and purchase decision support, readers should contact ECRI's SELECT™ Group.

Stage of development

The mid-1980s witnessed the introduction of microprocessor-based ventilators that could be easily upgraded to perform additional operations by a simple software change. However, the use of microprocessors has given the operator a vast and sometimes confusing number of options to choose from. Over the next several years, monitors for gas exchange and hemodynamics may be merged with the ventilator's existing data collection system. This combined system may alert the clinician to necessary control changes.

Recently, the concept of tracheal triggering was introduced. Tracheal pressure triggering can substantially reduce the work of breathing in lung models stimulating spontaneous breathing with CPAP. This reduction occurs because a small level of pressure support is produced at the proximal endotracheal tube. Tracheal triggering may also be beneficial when small endotracheal tubes are used.

There has been a recent trend to collect and store information from medical devices electronically and to display information from one device on another. The Institute of Electrical and Electronics Engineers has been developing a standard communication protocol called the medical information bus, which allows many types of medical devices to communicate with each other and to transfer data in a standardized format. Institutions with more than one brand of ventilator may thereby be able to automate their respiratory care charting.

In addition, most ventilators now offer a noninvasive delivery option. This option delivers ventilation usually through a mask that fits over the mouth and nose.

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Ventilators, Intensive Care

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The Netherlands
Phone: 31 (33) 2535404 Fax: 31 (33) 2537223
Internet: <http://www.datex-ohmeda.com>
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Datex-Ohmeda Inc, Div GE Healthcare [351254]
3030 Ohmeda Dr PO Box 7550
Madison, WI 53707-7550
Phone: (608) 221-1551 (800) 345-2700 Fax: (608) 222-9147
Internet: <http://www.us.datex-ohmeda.com>
E-mail: info@us.datex-ohmeda.com

Datex-Ohmeda KK [370383]
TRC Annex 9/F1 6-1-1 Hiewajima Ohta-ku
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Japan
Phone: 81 (3) 57636801 Fax: 81 (3) 57636838
Internet: <http://www.datex-ohmeda.com>
E-mail: kivokazu.kanai@jp.datex-ohmeda.com

Hamilton

Hamilton Medical AG [138228]
via Nova
CH-7403 Rhaezuens
Switzerland
Phone: 41 (81) 6606010 Fax: 41 (81) 6606020
Internet: <http://www.hamilton-medical.ch>
E-mail: ms@hamilton-medical.ch

Hamilton Medical Asia Pacific [418184]
Temasek Avenue 1 #27-01 Millania Tower
Singapore 039192
Republic of Singapore
Phone: 65 63569541 Fax: 65 63531673
Internet: <http://www.hamilton-medical.com>
E-mail: marketing@hammed1.com

Hamilton Medical Inc [105689]
4990 Energy Way PO Box 30008
Reno, NV 89520-3008
Phone: (775) 858-3200 (800) 426-6331 Fax: (775) 856-5621
Internet: <http://www.hamilton-medical.com>
E-mail: marketing@hammed1.com

Impact

Impact Instrumentation Inc [101885]
27 Fairfield Pl PO Box 508
West Caldwell, NJ 07006-0508
Phone: (973) 882-1212 (800) 969-0750 Fax: (973) 882-4993
Internet: <http://www.impactinstrumentation.com>
E-mail: info@impactinstrumentation.com

Intermed

Intermed Equipamento Medico Hospitalar Ltda [174394]
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Sao Paulo-SP 04366-000
Brazil
Phone: 55 (11) 56701303 Fax: 55 (11) 55624862

Ventilators, Intensive Care

Internet: <http://www.intermed.com.br>

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Kimura

S Kimura Medical Instrument Co Ltd [152416]

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Tokyo 113

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Phone: 81 (3) 38144061 Fax: 81 (3) 38145304

Internet: <http://www.kimura-medical.co.jp>

E-mail: tokyo_intl@kimura-medical.co.jp

Maquet

Maquet Critical Care AB, A Getinge Group Co [439169]

Rontgenvagen 2

S-171 95 Solna

Sweden

Phone: 46 (8) 7307300 Fax: 46 (8) 985775

Internet: <http://www.maquet.com/criticalcare>

E-mail: anna.khalil@maquet.com

Maquet GmbH & Co KG, A Getinge Group Co [305482]

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Phone: 49 (7222) 9320 Fax: 49 (7222) 932828

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E-mail: info.sales@maquet.de

Maquet Inc, A Getinge Group Co [336117]

1140 E Route 22 Suite 202

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Phone: (888) 627-8383 Fax: (908) 947-2301

Internet: <http://www.maquet-inc.com>

E-mail: customer.support@maquet-inc.com

Newport

Newport Medical Instruments Europe [187402]

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Barton-upon-Humber South Humberside DN18 5HN

England

Phone: 44 (4682) 31311

Internet: <http://www.ventilators.com>

E-mail: nasales@newportnmi.com

Newport Medical Instruments Inc [105093]

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Newport Beach, CA 92658

Phone: (714) 427-5811 (800) 451-3111 Fax: (714) 427-0489

Internet: <http://www.ventilators.com>

E-mail: customers@ventilators.com

Pulmonetic Systems/VIASYS Healthcare

Pulmonetic Systems Inc, VIASYS Healthcare Respiratory Technology [366511]

17400 Medina Rd Suite 100

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Internet: <http://www.pulmonetic.com>

E-mail: info@pulmonetic.com

Respironics

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Tokyo 112-0034
Japan
Phone: 81 (3) 58000724 Fax: 81 (3) 58000722
Internet: <http://www.respironics.com>
E-mail: sales@respironics.com

Respironics France [321323]
EAME 11 rue de Paris
F-92100 Boulogne Billancourt
France
Phone: 33 (1) 55601980 Fax: 33 (1) 55601989
Internet: <http://www.respironics.com>
E-mail: sales@respironics.com

Respironics Inc [101639]
1010 Murry Ridge Ln
Murrysville, PA 15668-8525
Phone: (724) 387-5200 (800) 345-6443 Fax: (724) 387-5010
Internet: <http://www.respironics.com>
E-mail: sales@respironics.com

Saime

Saime SA [263810]
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France
Phone: 33 (1) 64191111 Fax: 33 (1) 64418130
Internet: <http://www.saime.fr>
E-mail: jlroux@saime.net

Siare

Siare Hospital Supplies srl [152520]
via Giulio Pastore 18
I-40056 Crespellano BO
Italy
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Taema

Taema, Sub L'Air Liquide SA [151544]
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France
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Internet: <http://www.taema.com>
E-mail: olivier.tessier@airliquide.com

Tecme

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Argentina
Phone: 54 (351) 4651067 Fax: 54 (351) 4650208
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Tyco Healthcare Puritan Bennett

Puritan Bennett Corp, Div Tyco Healthcare Group LP [101913]
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Tyco Healthcare Pte Ltd, Div Tyco Healthcare Group LP [399028]
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Singapore 569507
Republic of Singapore
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Internet: <http://www.tycohealthcare.com>
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Tyco Healthcare UK Ltd, Div Tyco Healthcare Group LP [398199]
154 Fareham Road
Gosport Hampshire PO13 OAS
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Phone: 44 (1329) 224000 Fax: 44 (1329) 220213
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VersaMed

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VIASYS Healthcare

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VIASYS Healthcare Inc, Critical Care Div [444110]
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Palm Springs, CA 92262

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About the chart specifications

The following terms are used in the chart:

I:E ratio: The ratio of inspiratory time to expiratory time.

PEEP/CPAP, cm H₂O: PEEP is positive end-expiratory pressure; CPAP is continuous positive airway pressure. Both are expressed in cm H₂O.

Assist mode: The ventilator delivers breaths when it senses the patient's inspiratory efforts.

Control mode: The ventilator delivers breaths at chosen intervals.

IMV/SIMV: Intermittent mandatory ventilation/ synchronized intermittent mandatory ventilation.

Abbreviations:

The following abbreviations are used in the chart:

ANSI — American National Standards Institute

APRV — Airway pressure release ventilation

ARO — After receipt of order

BGM — Bird graphics monitor

BTPS — Body temperature pressure saturation

CE mark — Conformance Europeene mark

CEI — Comitato Elettrotecnico Italiano

cETL — Canadian ETL Testing Laboratories

CISPR — Comite International Special des Perturbations Radioelectrique (International Special Committee on Radio Interference)

CMV — Controlled mechanical ventilation

CPAP — Continuous positive airway pressure

CPU — Central processing unit

CSA — Canadian Standards Association

DIN — Deutsches Institut fuer Normung

EL — Electroluminescent

EN — European Norm

FDA — U.S. Food and Drug Administration

FiO₂ — Fraction of inspired oxygen

GLEM — Groupement des Laboratoires d'Essais des Matériels de Technique Medicale

I:E ratio — Inspiratory:expiratory ratio

IEC — International Electrotechnical Commission

ISO — International Organization for Standardization

JIS — Japanese Industrial Standards

JMMI — Japanese Machine and Metals Institute

kPa — Kilopascals

LCD — Liquid crystal display

LED — Light-emitting diode

MAP — Mean airway pressure

MDD — Medical Devices Directive

MIP — Mean inspiratory pressure

MMV — Mandatory minute volume

MV — Minute volume (liters/minute, or L/min)

NFC — National Fire Code

Ni-Cd — Nickel-cadmium

Ni-MH — Nickel-metal hydride

NRTL — Nationally Recognized Testing Laboratory

PC — Personal computer

PCV — Pressure-controlled ventilation
PEEP — Positive end-expiratory pressure
PIP — Peak inspiratory pressure
PRVC — Pressure-regulated volume control
psig — Pounds per square inch gauge
PSV — Pressure-support ventilation
RSBI — Rapid shallow breathing index
SEV — Schweizerischer Electrotechnischer Verein
SIMV — Synchronized intermittent mandatory ventilation
SLA — Sealed lead-acid

SVGA — Super Video Graphics Array
TFT — Thin-film transistor
TUV — Technischer Ueberwachungs Verein
TV — Tidal volume
UL — Underwriters Laboratories, Inc
UPS — Uninterruptible power supply
VAPS — Volume-assured pressure support
VCV — Volume-controlled ventilation
VGA — Video Graphics Array
WOBimp — Imposed work of breathing

Note: The data in the charts derive from suppliers' specifications and have not been verified through independent testing by ECRI or any other agency. Because test methods vary, different products' specifications are not always comparable. Moreover, products and specifications are subject to frequent changes. ECRI is not responsible for the quality or validity of the information presented or for any adverse consequences of acting on such information.

When reading the charts, keep in mind that, unless otherwise noted, the list price does not reflect supplier discounts. And although we try to indicate which features and characteristics are standard and which are not, some may be optional, at additional cost.

For those models whose prices were supplied to us in currencies other than U.S. dollars, we have also listed the conversion to U.S. dollars *to facilitate comparison among models*. However, keep in mind that exchange rates change often.

Need to know more?

For further information about the contents of this Product Comparison, contact the *HPCS* Hotline at +1 (610) 825-6000, ext. 5265; +1 (610) 834-1275 (fax); or hpcs@ecri.org (e-mail).

Product Comparison Chart

Product Comparison Chart

MODEL	ECRI-RECOMMENDED SPECIFICATIONS ¹ Basic IC Ventilator	ECRI-RECOMMENDED SPECIFICATIONS ¹ Mid/High IC Ventilator	ACOMA ART-1000	ACOMA ART-21EX
WHERE MARKETED			Not specified	Not specified
FDA CLEARANCE			Not specified	Not specified
CE MARK (MDD)			Not specified	Not specified
PATIENT TYPE			Not specified	Not specified
CONTROLS				
Tidal volume, mL	50-800	50-800	50-1,300	50-1,300
Inspiratory flow, L/min	3-180	3-180	5-65	5-65
Inspiratory pressure, cm H ₂ O	0-80	5-60	5-70	0-70
Respiratory rate, breaths/min	6-120	6-120	6-40	6-40
Inspiratory time, sec	0-3 pause	0-3 pause	0.5-5 (SIMV)	0.5-5
Expiratory time, sec	1-8	1-8	Not specified	Not specified
IE ratio	1:4 to 4:1	1:4 to 4:1	1:0.3 to 1:3	1:0.3 to 1:3
Inspiratory hold/plateau	0-3 sec	0-3 sec	0-50% inspiratory time	0-30% inspiratory time
Expiratory hold	0-3 sec	0-3 sec	Not specified	Not specified
FiO ₂ , %	30-90	30-90	21-100	21-100
Manual breath	Yes	Yes	No	Yes
PEEP/CPAP, cm H ₂ O	0-45	0-45	0-20	0-20
Pressure support, cm H ₂ O	0-45	0-45	Not specified	Not specified
Nebulizer	Optional	Optional	Yes	Yes
Trigger mechanism	Pressure, flow, both	Pressure, flow, both	Pressure, flow	Flow
Bias/base flow range, L/min	1-20	1-20	10-30	5
Pressure slope/ramp adjustment	Yes/yes	Yes/yes	Not specified	NA
Sigh	Optional	Optional	Yes	Yes
100% O ₂			Not specified	Not specified
Others			None specified	None specified

This is the first of four pages covering the above model(s). These specifications continue onto the next three pages.

Product Comparison Chart

MODEL	ECRI-RECOMMENDED SPECIFICATIONS ¹ Basic IC Ventilator	ECRI-RECOMMENDED SPECIFICATIONS ¹ Mid/High IC Ventilator	ACOMA ART-1000	ACOMA ART-21EX
OPERATING MODES				
Assist/control				
Volume breaths	Yes	Yes	Yes	Yes
Pressure breaths	Yes	Yes	Yes	Yes
SIMV				
Volume breaths	Yes	Yes	Yes	Yes
Pressure breaths	Optional	Yes	Yes	Yes
Pressure support	Yes	Yes	Yes	Yes
Spontaneous/CPAP				
Pressure support	Yes	Yes	Yes	Yes
Apnea-backup vent	Yes	Yes	Yes	Yes
Combination modes		Yes	Not specified	Not specified
Responsive valve		Yes	Not specified	Not specified
Bilevel/APRV		Yes	Not specified	Not specified
Others	User preference	User preference	Pressure-limited CMV	None specified
MONITORED PARAMETERS				
Pressure				
PIP	Yes	Yes	Yes	Yes
MAP	Yes	Yes	Yes	Yes
PEEP	Yes	Yes	Yes	Yes
Volume				
Tidal	Yes	Yes	Yes	Yes
Minute	Yes	Yes	Yes	Yes
Spontaneous minute	Optional	Yes	Yes	Yes
FiO2	Yes	Yes	Optional	Optional
Respiratory rate	Yes	Yes	Not specified	Yes
Inspiratory time	Yes	Yes	Yes	Yes
Expiratory time	Yes	Yes	Yes	Yes
IE	Yes	Yes	Yes	Yes
Others	Based on user requirements	Based on user requirements	None specified	None specified
PATIENT ALARMS				
FiO2	Yes	Yes	Yes	Yes
Low minute volume	Yes	Yes	Yes	Yes
Low inspiratory pressure	Yes	Yes	Yes	Yes
High pressure	Yes	Yes	Yes	Yes
Loss of PEEP	Optional	Yes	Yes	Yes
Apnea	Yes	Yes	Yes	Yes
High continuous pressure/occlusion	Yes	Yes	Not specified	Not specified
Inverse IE	Yes	Yes	Not specified	Not specified
High respiratory rate	Yes	Yes	Yes	Yes
High minute volume	Optional	Yes	Yes	Yes
High PEEP	Optional	Yes	Yes	Yes
Breathing circuit disconnect			Not specified	Not specified
Others			None specified	None specified

This is the second of four pages covering the above model(s). These specifications continue onto the next two pages.

Product Comparison Chart

MODEL	ECRI-RECOMMENDED SPECIFICATIONS ¹ Basic IC Ventilator	ECRI-RECOMMENDED SPECIFICATIONS ¹ Mid/High IC Ventilator	ACOMA ART-1000	ACOMA ART-21EX
EQUIPMENT ALARMS				
Gas supply failure	Yes	Yes	Yes	Yes
Power failure	Yes	Yes	Yes	Yes
Vent inoperative	Yes	Yes	Yes	Yes
Low battery	Yes	Yes	Not specified	Yes
Self-diagnostics	Valve leak, sensor failure	Valve leak, sensor failure	CPU error	CPU error
Others	By user requirements	By user requirements	Flow detector connection, apnea, flow rate, preset failure	Flow rate, preset failure
INTERFACING				
Output ports	Optional	Optional	No	No
Remote alarm/display	Optional	Optional	No	No
Analog output	Optional	Optional	Optional	No
Report generation				
Display	Required	Required	No	Not specified
Hard copy	Required	Required	No	Not specified
Archival disk	Preferred	Preferred	No	Not specified
Network	Preferred	Preferred	No	Not specified
DISPLAY TYPES	User preference	User preference	LED	LED
DATA DISPLAYED	User customizable	User customizable	Not specified	Not specified
PNEUMATIC POWER				
Compressor	Optional	Optional	Not necessary	Optional
Compressed gases	O2, air	O2, air	O2	Not specified
Pressure ranges	35-65 psi	35-65 psi	≤250 kPa	20 kPa
Turbine/piston			Not specified	Not specified
LINE POWER, VAC	Standard	Standard	100-240, 50/60 Hz	100-240, 50/60 Hz
Current, amps			Not specified	Not specified
Watts			300	330
INTERNAL BATTERY	Required	Required	No	Yes
Type (number)	Any common type	Any common type	NA	Not specified
Operating time, hr	1	1	NA	Not specified
Rechargeable			NA	Not specified
Recharging time, hr			NA	Not specified
EXTERNAL BATTERY			Not specified	Not specified
Type (number)			Not specified	Not specified
Operating time, hr			Not specified	Not specified
Rechargeable			Not specified	Not specified
Recharging time, hr			Not specified	Not specified
H x W x D, cm (in)			136.4 x 49.4 x 45 (53.7 x 19.5 x 17.7)	125 x 40 x 62 (50 x 15.7 x 24.8)
WEIGHT, kg (lb)			70 (154.3)	67 (148)

This is the third of four pages covering the above model(s). These specifications continue onto the next page.

Product Comparison Chart

MODEL	ECRI-RECOMMENDED SPECIFICATIONS ¹ Basic IC Ventilator	ECRI-RECOMMENDED SPECIFICATIONS ¹ Mid/High IC Ventilator	ACOMA ART-1000	ACOMA ART-21EX
PURCHASE INFORMATION				
List price			\$20,000	Not specified
Warranty			1 year	1 year
Service				
Factory/on-site			Not specified	Not specified
Training			Not specified	Not specified
3rd-party service			Not specified	Not specified
Parts availability			Not specified	Not specified
Delivery time, ARO			Not specified	Not specified
Year first sold			Not specified	Not specified
Number sold				
USA/worldwide			Not specified	Not specified
Fiscal year			Not specified	Not specified
OTHER			None specified.	None specified.
SPECIFICATIONS				
LAST UPDATED			April 2006	April 2006
Supplier Footnotes	¹ These recommendations are the opinions of ECRI's technology experts. ECRI assumes no liability for decisions made based on this data.	¹ These recommendations are the opinions of ECRI's technology experts. ECRI assumes no liability for decisions made based on this data.		
Model Footnotes				
Data Footnotes				

Product Comparison Chart

MODEL	BIO-MED DEVICES CV-3	BIO-MED DEVICES CV-4	DRAEGER Carina home	DRAEGER Evita 2 dura
WHERE MARKETED	Worldwide	Worldwide	Worldwide	Worldwide
FDA CLEARANCE	Yes	Yes	No	Yes
CE MARK (MDD)	Yes	Yes	Yes	Yes
PATIENT TYPE	Adult to infant	Adult to neonatal	Not specified	Adult, pediatric, neonatal
CONTROLS				
Tidal volume, mL	5-2,500	5-2,500	100-2,000	3-2,000 with NeoFlow
Inspiratory flow, L/min	1-120	1-120	NA	6-180
Inspiratory pressure, cm H ₂ O	0-120	0-120	5-50	0-80
Respiratory rate, breaths/min	5-150	5-150	0-50	0-150
Inspiratory time, sec	0.1-3	0.1-3	5-50	0.1-30
Expiratory time, sec	0.2-120	0.2-120	NA	0.1-30
IE ratio	3:1 to 1:99	3:1 to 1:99	1:3 to 2:1	1:300 to 300:1
Inspiratory hold/plateau	No	0-1/3 inspiratory time	No	Yes
Expiratory hold	Not specified	Not specified	No	Yes
FiO ₂ , %	21-100 with optional blender	21-100 with optional blender	NA	21-100
Manual breath	Yes	Yes	Yes	Yes
PEEP/CPAP, cm H ₂ O	0-35	0-35	3-20	0-35
Pressure support, cm H ₂ O	0-50	0-50	3-40	0-80
Nebulizer	No	Yes	Yes	Yes
Trigger mechanism	Pressure	Pressure	Flow change, volume, pressure	Flow (pressure)
Bias/base flow range, L/min	Not specified	Not specified	Not specified	Not specified
Pressure slope/ramp adjustment	Not specified	Not specified	Yes	Yes/yes
Sigh	Yes	Yes	No	Yes
100% O ₂	Not specified	Not specified	No	Yes
Others	None specified	None specified	None specified	None specified

This is the first of four pages covering the above model(s). These specifications continue onto the next three pages.

Product Comparison Chart

MODEL	BIO-MED DEVICES CV-3	BIO-MED DEVICES CV-4	DRAEGER Carina home	DRAEGER Evita 2 dura
OPERATING MODES				
Assist/control				
Volume breaths	Yes	Yes	Yes	Yes, AutoFlow
Pressure breaths	Yes	Yes	Yes	Yes
SIMV				
Volume breaths	Yes	Yes	Yes	Yes
Pressure breaths	Yes	Yes	Yes	Yes
Pressure support	Yes	Yes	Yes	Yes
Spontaneous/CPAP				
Pressure support	Yes	Yes	Yes	Yes
Apnea-backup vent	Yes	Yes	Yes	Yes
Combination modes	Not specified	Not specified	Not specified	See Others
Responsive valve	Not specified	Not specified	Not specified	Yes
Bilevel/APRV	Not specified	Not specified	Not specified	Yes
Others	None specified	SIMV with shift delay	I:E ratio 1:3 to 2:1, LPO only, no oxygen control, 13 hr external battery extends battery operation time	APRV, MMV, AutoFlow, PCV+, automatic-tube compensation (all patient ranges) including nCPAP; all modes have noninvasive delivery option; optional independent lung ventilation
MONITORED PARAMETERS				
Pressure				
PIP	Yes	Yes	Yes	Yes
MAP	Yes	Yes	No	Yes
PEEP	Yes	Yes	No	Yes
Volume				
Tidal	Yes	Yes	Yes	Yes
Minute	Yes	Yes	No	Yes
Spontaneous minute	Yes	Yes	No	Yes
FiO2	Yes	Yes	No	Yes
Respiratory rate	Yes	Yes	Yes	Yes
Inspiratory time	Yes	Yes	No	Yes
Expiratory time	Yes	Yes	No	Yes
IE	Yes	Yes	No	Yes
Others	None specified	None specified	Not specified	Optional occlusion pressure, NIF, RSBI, capnogram, new sensor, no calibration data, remote fault, transducer fault, resistance, compliance, breathing gas temperature, leak rate for minute volume
PATIENT ALARMS				
FiO2	Yes	Yes	No	Yes
Low minute volume	Yes	Yes	No	Yes
Low inspiratory pressure	Yes	Yes	No	Yes
High pressure	Yes	Yes	Yes	Yes
Loss of PEEP	Yes	Yes	No	Yes
Apnea	Yes	Yes	Yes	Yes
High continuous pressure/occlusion	Yes	Yes	Yes	Yes
Inverse IE	Yes	Yes	No	Yes
High respiratory rate	Yes	Yes	Yes	Yes
High minute volume	Yes	Yes	No	Yes
High PEEP	Yes	Yes	No	Yes
Breathing circuit disconnect	Yes	Yes	Yes	Yes
Others	None specified	None specified	High tidal volume	High tidal volume

This is the second of four pages covering the above model(s). These specifications continue onto the next two pages.

Product Comparison Chart

MODEL	BIO-MED DEVICES CV-3	BIO-MED DEVICES CV-4	DRAEGER Carina home	DRAEGER Evita 2 dura
EQUIPMENT ALARMS				
Gas supply failure	Yes	Yes	Yes	Yes
Power failure	Yes	Yes	Yes	Yes
Vent inoperative	Yes	Yes	Yes	Yes
Low battery	Yes	Yes	Yes	Yes
Self-diagnostics	Yes	Yes	Yes	Yes
Others	None specified	None specified	Wrong patient system, rescue mode	Exhalation valve, flow sensor insertion, leak, compliance
INTERFACING				
Output ports	RS232	RS232	RS232, analog	RS232, analog
Remote alarm/display	Not specified	Optional	Optional	Optional
Analog output	Not specified	No	Yes	Yes
Report generation				
Display	Not specified	Yes	Not specified	Via VentView
Hard copy	Not specified	Optional	Not specified	Via VentView
Archival disk	Not specified	Optional	Not specified	Via VentView
Network	Not specified	Optional	Not specified	Via VentView
DISPLAY TYPES	LCD with backlit touchscreen	LCD with backlit touchscreen	TFT BW	TFT color screen
DATA DISPLAYED	Not specified	Not specified	Numbers, waveforms	Numbers, waveforms, loops, and trends
PNEUMATIC POWER				
Compressor	Available	Available	Not specified	Optional
Compressed gases	O2, air	O2, air	O2, air	O2, air
Pressure ranges	44-66 psig	44-66 psig	500 hPa maximum	2.7-6 bar, 40-87 psi
Turbine/piston	Not specified	Not specified	Turbine	Not specified
LINE POWER, VAC	110/220	110/220	100-127/220-241	100-127/220-240
Current, amps	Not specified	Not specified	Not specified	1.3 (230 VAC) maximum, 3.2 (100 VAC) maximum
Watts	Not specified	Not specified	Not specified	125
INTERNAL BATTERY	Yes	Yes	Yes	Optional
Type (number)	Not specified	Not specified	Internal (not specified)	Internal (not specified)
Operating time, hr	10	10	2	2
Rechargeable	Yes	Yes	Yes	Yes
Recharging time, hr	5	5	2	Not specified
EXTERNAL BATTERY	Not specified	Not specified	Yes	Yes
Type (number)	Not specified	Not specified	Not specified (2)	Not specified (2)
Operating time, hr	Not specified	Not specified	13	Not specified
Rechargeable	Not specified	Not specified	Yes	Not specified
Recharging time, hr	Not specified	Not specified	8	Not specified
H x W x D, cm (in)	27.9 x 24.8 x 13.7 (11 x 9.8 x 5.4)	27.9 x 24.8 x 13.7 (11 x 9.8 x 5.4)	17.5 x 27.5 x 38.5 (6.9 x 10.8 x 15.2)	53 x 29 x 45 (20.9 x 11.4 x 17.7); 58 x 131.5 x 66 (22.8 x 51.8 x 26) with trolley
WEIGHT, kg (lb)	4.1 (9)	4.1 (9)	4.9 (10.8)	27 (59.5); 69 (152) with trolley and cabinet

This is the third of four pages covering the above model(s). These specifications continue onto the next page.

Product Comparison Chart

MODEL	BIO-MED DEVICES CV-3	BIO-MED DEVICES CV-4	DRAEGER Carina home	DRAEGER Evita 2 dura
PURCHASE INFORMATION				
List price	Not specified	Not specified	\$6,500	\$25,550
Warranty	1 year	1 year	1 year	1 year
Service				
Factory/on-site	Yes/yes	Yes/yes	Yes/yes	Yes/yes
Training	Yes	Yes	Yes	Yes
3rd-party service	Not specified	No	Yes	Yes
Parts availability	Yes	Yes	Yes	Yes
Delivery time, ARO	Not specified	Not specified	4-6 weeks	4 weeks
Year first sold	1997	1995	2006	1997
Number sold				
USA/worldwide	Not specified	Not specified	Not specified	Not specified
Fiscal year	October to September	October to September	January to December	January to December
OTHER SPECIFICATIONS	None specified.	Auto patient-compliance calculations; auto alarm settings; user help screens; foreign-language menus; 12-30 V input jack for transport; infant menu with inspiratory time and constant flow; optional air entrainment for 55-60% O ₂ concentration.	All gas measurements under BTPS conditions. Meets requirements of IEC 60601.	All gas measurements under BTPS conditions; single rotary knob parameter adjustment; options and upgrades via software and/or hardware; tubing system compliance compensated; open breathing system in all modes and patient ranges; optional SBCO ₂ (single-breath CO ₂) and SpO ₂ . Meets requirements of IEC 60601.
LAST UPDATED	April 2006	April 2006	April 2006	April 2006
Supplier Footnotes				
Model Footnotes				
Data Footnotes				

Product Comparison Chart

MODEL	DRAEGER Evita 4	DRAEGER Evita XL	DRAEGER Oxylog 1000	DRAEGER Oxylog 2000
WHERE MARKETED	Worldwide	Worldwide	Worldwide	Worldwide
FDA CLEARANCE	Yes	Yes	Yes	Yes
CE MARK (MDD)	Yes	Yes	Yes	Yes
PATIENT TYPE	Adult, pediatric, neonatal	Adult, pediatric, neonatal	Not specified	Not specified
CONTROLS				
Tidal volume, mL	3-2,000 with NeoFlow	3-2,000 with NeoFlow	Not specified	100-1,500
Inspiratory flow, L/min	6-180	6-180	3-20	4-60
Inspiratory pressure, cm H ₂ O	0-80	0-95	25-55	20-60
Respiratory rate, breaths/min	0-150	0-300	4-54	5-40
Inspiratory time, sec	0.1-30	0.1-30	Depends on set frequency	Depends on set frequency and I:E ratio
Expiratory time, sec	0.1-30	0.1-30	Depends on set frequency	Depends on set frequency and I:E ratio
IE ratio	1:300 to 300:1	1:300 to 300:1	1:1.5	1:3 to 2:1
Inspiratory hold/plateau	Yes	Yes	Not specified	Not specified
Expiratory hold	Yes	Yes	Not specified	Not specified
FiO ₂ , %	21-100	21-100	60 or 100	60 or 100
Manual breath	Yes	Yes	No	No
PEEP/CPAP, cm H ₂ O	0-35	0-50	0-20	0-15
Pressure support, cm H ₂ O	0-80	0-80	Not specified	Not specified
Nebulizer	Yes	Yes	Not specified	Not specified
Trigger mechanism	Flow (pressure)	Flow (pressure)	NA	Flow
Bias/base flow range, L/min	Not specified	Not specified	Not specified	Not specified
Pressure slope/ramp adjustment	Yes/yes	Yes/yes	Not specified	Not specified
Sigh	Yes	Yes	Not specified	Not specified
100% O ₂	Yes	Yes	Yes	Yes
Others	None specified	None specified	CPR mode: inspiration phase not interrupted; more O ₂ exchange	CPR mode: inspiration phase not interrupted; more O ₂ exchange

This is the first of four pages covering the above model(s). These specifications continue onto the next three pages.

Product Comparison Chart

MODEL	DRAEGER Evita 4	DRAEGER Evita XL	DRAEGER Oxylog 1000	DRAEGER Oxylog 2000
OPERATING MODES				
Assist/control				
Volume breaths	Yes, AutoFlow	Yes, AutoFlow	Not specified	Yes
Pressure breaths	Yes	Yes	Not specified	Not specified
SIMV				
Volume breaths	Yes	Yes	Not specified	Yes
Pressure breaths	Yes	Yes	Not specified	Pressure limited
Pressure support	Yes	Yes	Not specified	Not specified
Spontaneous/CPAP				
Pressure support	Yes	Yes	Not specified	Not specified
Apnea-backup vent	Yes	Yes	NA	Not specified
Combination modes	See Others	See Others	Not specified	Not specified
Responsive valve	Yes	Yes	Not specified	Not specified
Bilevel/APRV	Yes	Yes	Not specified	Not specified
Others	APRV, MMV, AutoFlow, PCV+, automatic-tube compensation (all patient ranges) including nCPAP; all modes have noninvasive delivery option; optional independent lung ventilation	SmartCare (knowledge-based weaning system), APRV, MMV, AutoFlow, PCV+, automatic-tube compensation (all patient ranges) including nCPAP; all modes have noninvasive delivery option; optional independent lung ventilation	None specified	None specified
MONITORED PARAMETERS				
Pressure				
PIP	Yes	Yes	Yes	Yes
MAP	Yes	Yes	Not specified	Yes
PEEP	Yes	Yes	Not specified	Yes
Volume				
Tidal	Yes	Yes	Not specified	Yes
Minute	Yes	Yes	Not specified	Yes
Spontaneous minute	Yes	Yes	NA, CMV ventilation	Not specified
FiO2	Yes	Yes	Not specified	Not specified
Respiratory rate	Yes	Yes	Not specified	Yes
Inspiratory time	Yes	Yes	Not specified	Yes
Expiratory time	Yes	Yes	Not specified	Not specified
IE	Yes	Yes	Not specified	Not specified
Others	Optional occlusion pressure, NIF, RSBi, capnogram, new sensor, no calibration data, remote fault, transducer fault, resistance, compliance, breathing gas temperature, leak rate for minute volume	Optional occlusion pressure, NIF, RSBi, capnogram, new sensor, no calibration data, remote fault, transducer fault, resistance, compliance, breathing gas temperature, leak rate for minute volume	None specified	CPAP pressure level, Vt expiratory
PATIENT ALARMS				
FiO2	Yes	Yes	Not specified	Not specified
Low minute volume	Yes	Yes	Not specified	Yes
Low inspiratory pressure	Yes	Yes	Yes	Yes
High pressure	Yes	Yes	Yes	Yes
Loss of PEEP	Yes	Yes	Not specified	Not specified
Apnea	Yes	Yes	NA, CMV ventilation	Yes
High continuous pressure/occlusion	Yes	Yes	Yes	Yes
Inverse IE	Yes	Yes	Not specified	Not specified
High respiratory rate	Yes	Yes	Not specified	Yes
High minute volume	Yes	Yes	Not specified	Yes
High PEEP	Yes	Yes	Not specified	Not specified
Breathing circuit disconnect	Yes	Yes	Yes (Paw low alarm)	Yes
Others	High tidal volume	High tidal volume	None specified	None specified

These specifications
continue onto the next two
pages

Product Comparison Chart

MODEL	DRAEGER Evita 4	DRAEGER Evita XL	DRAEGER Oxylog 1000	DRAEGER Oxylog 2000
EQUIPMENT ALARMS				
Gas supply failure	Yes	Yes	Yes	Yes
Power failure	Yes	Yes	Yes (100% pneumatic)	Yes
Vent inoperative	Yes	Yes	NA (100% pneumatic)	Yes
Low battery	Yes	Yes	Yes; low pressure supply, no battery needed	Yes
Self-diagnostics	Yes	Yes	Not specified	Yes
Others	Exhalation valve, flow sensor insertion, leak, compliance	Exhalation valve, flow sensor insertion, leak, compliance	Audible and visual alarms	None specified
INTERFACING				
Output ports	RS232, analog	RS232, analog	NA (100% pneumatic)	Not specified
Remote alarm/display	Optional	Optional	NA (100% pneumatic)	Not specified
Analog output	Yes	Yes	NA (100% pneumatic)	Not specified
Report generation				
Display	Via VentView	Via VentView	NA (100% pneumatic)	Not specified
Hard copy	Via VentView	Via VentView	NA (100% pneumatic)	Not specified
Archival disk	Via VentView	Via VentView	NA (100% pneumatic)	Not specified
Network	Via VentView	Via VentView	NA (100% pneumatic)	Not specified
DISPLAY TYPES	TFT color touchscreen	TFT color touchscreen	None	4 x 20 LCD, 9 languages; adjustable
DATA DISPLAYED	Numbers, waveforms, loops, and trends	Numbers, waveforms, loops, trends, short trends	NA	Settings, alarms, measured values, pressure and flow values and alarms
PNEUMATIC POWER				
Compressor	Optional	Optional	Not specified/not integrated	Not specified/not integrated
Compressed gases	O2, air	O2, air	Medical grade air or O2	Medical grade air or O2
Pressure ranges	2.7-6 bar, 40-87 psi	2.7-6 bar, 40-87 psi	2.7-6 bar, 39-90 psi	2.7-6 bar, 39-90 psi
Turbine/piston	Not specified	Not specified	Not specified	Not specified
LINE POWER, VAC	100-127/220-240	100-127/220-240	NA (100% pneumatic)	100-240
Current, amps	1.3 (230 VAC) maximum, 3.2 (100 VAC) maximum	1.3 (230 VAC) maximum, 3.2 (100 VAC) maximum	NA (100% pneumatic)	0.7, maximum 1.6
Watts	125	125	NA (100% pneumatic)	2.8 for ventilation, 3.6 for battery charging
INTERNAL BATTERY				
Type (number)	Optional Internal (not specified)	Optional Internal (not specified)	NA (100% pneumatic) NA (100% pneumatic)	Yes Ni-Cd or alkaline manganese (not specified)
Operating time, hr	2	2	NA (100% pneumatic)	Up to 6 (Ni-Cd), up to 4 (alkaline manganese)
Rechargeable	Yes	Yes	NA (100% pneumatic)	Yes
Recharging time, hr	Not specified	Not specified	NA (100% pneumatic)	8 maximum
EXTERNAL BATTERY	Yes	Yes	NA (100% pneumatic)	Not specified
Type (number)	Not specified (2)	Not specified (2)	NA (100% pneumatic)	Not specified
Operating time, hr	Not specified	Not specified	NA (100% pneumatic)	Not specified
Rechargeable	Not specified	Not specified	NA (100% pneumatic)	Not specified
Recharging time, hr	Not specified	Not specified	NA (100% pneumatic)	Not specified
H x W x D, cm (in)	53 x 29 x 45 (20.9 x 11.4 x 17.7); 58 x 133.5 x 66 (22.8 x 52.5 x 26) with trolley	53 x 29 x 45 (20.9 x 11.4 x 17.7); 58 x 133.5 x 66 (22.8 x 52.5 x 26) with trolley	7.6 x 21.5 x 21.5 (3 x 8.5 x 8.5)	12.3 x 21.5 x 20.8 (4.5 x 8.5 x 8.2)
WEIGHT, kg (lb)	27 (59.5); 69 (152) with trolley and cabinet	27 (59.5); 69 (152) with trolley and cabinet	3.15 (6.5)	4.3 (8.5)

This is the third of four pages covering the above model(s). These specifications continue onto the next page.

Product Comparison Chart

MODEL	DRAEGER Evita 4	DRAEGER Evita XL	DRAEGER Oxylog 1000	DRAEGER Oxylog 2000
PURCHASE INFORMATION				
List price	\$31,130	\$36,230	Not specified	Not specified
Warranty	1 year	1 year	1 year	1 year
Service				
Factory/on-site	Yes/yes	Yes/yes	Yes/yes	Yes/yes
Training	Yes	Yes	Yes	Yes
3rd-party service	Yes	Yes	Yes	Yes
Parts availability	Yes	Yes	Yes	Yes
Delivery time, ARO	4 weeks	4 weeks	Not specified	Not specified
Year first sold	1996	2002	1997	1994
Number sold				
USA/worldwide	Not specified	Not specified	Not specified	Not specified
Fiscal year	January to December	January to December	January to December	January to December
OTHER SPECIFICATIONS	Evita 2 platform plus touchscreen, ideal body weight setting, and cursor in displayed graphics.	Evita 4 platform plus screen customization (vent check sheet, waveforms, displayed values), multiple graphic capabilities (short trends, up to 6 loops, combined cursor for simultaneous loop and wave analysis), and online information for modes and alarms; lung protection package includes slow flow inflection maneuver; lung recruitment trending and direct link for pressure changes in pressure control.	Transport ventilator; outdoor usable (temperature range -18 to +50°C [-4 to +122°F], relative humidity 15-95%, ambient pressure 700-1,100 hPa). Vibration tested according to MIL STD 810 F (method 514.5). Airworthiness according to RTCA D-160D (section 8) IPX4.	Transport ventilator; outdoor usable (temperature range -18 to +50°C [-4 to +122°F], relative humidity 30-95%, ambient pressure 600-1,200 hPa). Vibration tested according to MIL STD 810 F (method 514.5). Airworthiness according to RTCA D-160D (section 8) IPX4.
LAST UPDATED	April 2006	April 2006	April 2006	April 2006
Supplier Footnotes				
Model Footnotes				
Data Footnotes				

Product Comparison Chart

MODEL	DRAEGER	DRAEGER	eVENT MEDICAL	GE HEALTHCARE (DATEX-OHMEDA)
	Oxylog 3000	Savina	Inspiration : Inspiration LS	Centiva/5 (12" Screen)
WHERE MARKETED	Worldwide	Worldwide	Worldwide	Worldwide ¹
FDA CLEARANCE	Submitted (expected 2006)	Yes	Yes	Yes
CE MARK (MDD)	Yes	Yes	Yes	Yes
PATIENT TYPE	Not specified	Adult, pediatric, neonatal	Adult, pediatric, infant	Adult to pediatric, optional neonatal
CONTROLS				
Tidal volume, mL	50-2,000	50-2,000	0-2,000	20-2,000
Inspiratory flow, L/min	100 maximum	Not specified	0-120 mandatory	2-99.6, 180 maximum peak flow
Inspiratory pressure, cm H ₂ O	0-55	0-99	0-80	1-59
Respiratory rate, breaths/min	2-60	2-80	1-150	4-100
Inspiratory time, sec	0.2-10	0.2-10	0.1-10	0.06-13.6, set via rate and I:E ratio
Expiratory time, sec	Not specified	Not specified	Not specified	0.2-13.6, set via rate and I:E ratio
IE ratio	1:4 to 3:1	1:150 to 150:1	59.9:1 to 1:10	1:9 to 4:1, 9:1 in bilevel
Inspiratory hold/plateau	Yes	Yes	10-70%	Yes
Expiratory hold	No	No	Not specified	No
FiO ₂ , %	40-100	21-100	21-100	21-100
Manual breath	Yes	Yes	Yes	Yes
PEEP/CPAP, cm H ₂ O	0-20	0-35	Not specified	Off, 2-35
Pressure support, cm H ₂ O	0-35 above PEEP	0-70	0-80	0-59
Nebulizer	Not specified	Yes	Yes	Pneumatic, synchronized, compensated
Trigger mechanism	Flow	Flow (pressure)	Pressure, flow	Flow
Bias/base flow range, L/min	Not specified	Not specified	Autoset, 2 above flow sensors	3-30
Pressure slope/ramp adjustment	Yes	Yes/yes	Fast, medium, slow	Rise time adjustment for inspiratory pressure and pressure support
Sigh	Not specified	Yes	Programmable, pressure and volume ventilation	No
100% O ₂	Yes	Yes	Yes	Yes, suction maneuver
Others	O ₂ inhalation (0-15 L/min O ₂)	5-200 mbar/sec flow acceleration	None specified	None specified

This is the first of four pages covering the above model(s). These specifications continue onto the next three pages.

Product Comparison Chart

MODEL	DRAEGER	DRAEGER	eVENT MEDICAL	GE HEALTHCARE (DATEX-OHMEDA)
	Oxylog 3000	Savina	Inspiration : Inspiration LS	Centiva/5 (12" Screen)
OPERATING MODES				
Assist/control				
Volume breaths	Yes	Yes, AutoFlow	Yes	Yes
Pressure breaths	Not specified	Yes	Yes	Bilevel
SIMV				
Volume breaths	Yes	Yes	Yes	Yes
Pressure breaths	Yes	Yes	Yes	Bilevel
Pressure support	Yes	Yes	Yes	Yes
Spontaneous/CPAP				
Pressure support	Yes	Yes	Yes	Yes
Apnea-backup vent	Yes	Yes	Yes	Yes
Combination modes	Not specified	See Others	Not specified	Yes
Responsive valve	Not specified	Yes	Not specified	Not specified
Bilevel/APRV	Not specified	No	Not specified	Yes
Others	PCV+, noninvasive delivery possible for CPAP and PCV+	AutoFlow, PCV+, nCPAP, all modes have noninvasive delivery option	SPAP, automode, PRVC, volume support, nCPAP	Bilevel, bilevel with Volume Guarantee ¹
MONITORED PARAMETERS				
Pressure				
PIP	Yes	Yes	Yes	Yes
MAP	Yes	Yes	Yes	Yes
PEEP	Yes	Yes	Yes	Yes
Volume				
Tidal	Yes	Yes	Yes	Yes
Minute	Yes	Yes	Yes	Yes
Spontaneous minute	Yes	Yes	Yes	Yes
FiO2	Yes	Yes	Yes	Yes
Respiratory rate	Yes	Yes	Yes	Yes
Inspiratory time	Not specified	Yes	No	No
Expiratory time	Not specified	Yes	No	No
IE	Not specified	Yes	Yes	No
Others	Plateau pressure, spontaneous respiratory rate	Plateau time, compliance, resistance, breathing gas temperature, leak rate for minute volume	Spontaneous rate, Ti, Ttot, RSBI, CSTAT, RES, spontaneous lh, spontaneous 8h, Pplat, % leak, high-low ratio, PEEP, V+I, I:E	Compliance, resistance, Pmin, RSBI, PEEPi, 12-day data, ventilator settings and alarm settings trends, alarm and event log
PATIENT ALARMS				
FiO2	Yes	Yes	Yes	Yes
Low minute volume	Yes	Yes	Yes	Yes
Low inspiratory pressure	Yes	Yes	Yes	Yes
High pressure	Yes	Yes	Yes	Yes
Loss of PEEP	Yes	Yes	No	Yes, indirectly
Apnea	Yes	Yes	Yes	Yes
High continuous pressure/occlusion	Yes	Yes	Yes	Yes
Inverse IE	Not specified	Not specified	Yes	Visual
High respiratory rate	Yes	Yes	Yes	Yes
High minute volume	Yes	Yes	No	Yes
High PEEP	Not specified	Yes	No	No
Breathing circuit disconnect	Yes	Yes	Yes	Yes
Others	None specified	High tidal volume	None specified	Low respiratory rate

This is the second of four pages covering the above model(s). These specifications continue onto the next two pages.

Product Comparison Chart

MODEL	DRAEGER	DRAEGER	eVENT MEDICAL	GE HEALTHCARE (DATEX-OHMEDA)
	Oxylog 3000	Savina	Inspiration : Inspiration LS	Centiva/5 (12" Screen)
EQUIPMENT ALARMS				
Gas supply failure	Yes	Yes	Yes	Yes
Power failure	Yes	Yes	Yes	Yes
Vent inoperative	Yes	Yes	Yes	Yes
Low battery	Yes	Yes	Yes	Yes
Self-diagnostics	Yes	Yes	Automatic calibrations and extensive logging of technical faults	Yes
Others	Leakage, flow sensor	Exhalation valve, flow sensor insertion, leak	None specified	Flow sensor, circuit leak, patient connect, patient disconnect
INTERFACING				
Output ports	IrDA RS232	RS232, analog	RS232, Ethernet	RS232
Remote alarm/display	Not specified	Optional	Yes	Yes, nurse call
Analog output	Not specified	No	MINIWEB	No
Report generation				
Display	Integrated EL display	Via optional graphic screen	MINIWEB	Via N-DISVENT
Hard copy	Not specified	Via optional graphic screen	MINIWEB	Via N-DISVENT
Archival disk	Via Customer Service Mode	Via optional graphic screen	MINIWEB	Via RS232 to external PC
Network	Not specified	Via optional graphic screen	MINIWEB	No
DISPLAY TYPES				
	EL; 9 languages adjustable	Black & white LCD	Graphical user interface	30.5 cm (12") diagonal color touchscreen LCD
DATA DISPLAYED				
	Settings, alarms, measured values, pressure and flow waveforms, stored events and alarms	Numbers, waveforms, bargraph	Not specified	Real-time pressure and flow waveforms, measured numerics, loops, settings, status
PNEUMATIC POWER				
Compressor	Not specified/not integrated	Built-in blower (turbine)	Emergency internal backup	Optional
Compressed gases	O2	O2	O2, air, Heliox (outside USA)	O2, air
Pressure ranges	2.7-6 bar, 39-90 psi	2.7-6 bar, 40-87 psi	2-6 bar	40-80 psi, 280-600 kPa
Turbine/piston	Not specified	Yes/no	Not specified	Not specified
LINE POWER, VAC				
	100-240	100-240	100-240, 50/60 Hz	98-132, 207-253; 50/60 Hz
Current, amps	2.1, maximum 3.8	1.3 (230 VAC) maximum, 3.4 (100 VAC) maximum	Not specified	Not specified
Watts	40	75	100	<50
INTERNAL BATTERY				
Type (number)	Yes Lithium ion or Ni-MH (not specified)	Standard Lead gel (not specified)	Yes Lead-acid (2)	Yes Sealed lead-acid (2)
Operating time, hr	~4 (lithium ion), 3 (Ni-MH)	1	5, 1.5 with internal compressor	0.5
Rechargeable	Yes	Yes	Yes	Yes
Recharging time, hr	~5 (lithium ion), 4 (Ni-MH)	2	5	12 (full discharge)
EXTERNAL BATTERY				
Type (number)	Not specified	Yes Lead or lead gel (not specified)	Not specified	Yes Sealed lead-acid (2) in special battery pack
Operating time, hr	Not specified	4	Not specified	1
Rechargeable	Not specified	Yes	Not specified	Yes
Recharging time, hr	Not specified	15	Not specified	10
H x W x D, cm (in)	18.4 x 28.5 x 17.5 (7.2 x 11.1 x 6.9)	38 x 38 x 36 (15 x 15 x 14), 137 x 55 x 56 (53 x 22 x 22) with trolley	48.3 x 33 x 38.1 (19 x 13 x 15) : 53.3 x 40.6 x 40.6 (21 x 16 x 16)	32 x 32 x 32 (12.6 x 12.6 x 12.6) CMS, 25 x 24 (9.8 x 9.4) footprint
WEIGHT, kg (lb)	5.4 (11.9)	24 (53)	21.8 (48) : 24 (53)	14.7 (32.3)

This is the third of four pages covering the above model(s). These specifications continue onto the next page.

Product Comparison Chart

MODEL	DRAEGER	DRAEGER	eVENT MEDICAL	GE HEALTHCARE (DATEX-OHMEDA)
	Oxylog 3000	Savina	Inspiration : Inspiration LS	Centiva/5 (12" Screen)
PURCHASE INFORMATION				
List price	Not specified	\$13,699	\$22,995	Not specified
Warranty	1 year	1 year	5 years, parts	1 year, parts and labor
Service				
Factory/on-site	Yes/yes	Yes/yes	Yes/yes	Yes/yes
Training	Yes	Yes	Yes	Yes
3rd-party service	Yes	Yes	Yes	Yes
Parts availability	Yes	Yes	Yes	Yes
Delivery time, ARO	Not specified	4 weeks	2 weeks	30-45 days
Year first sold	2003	2000	2001	2005
Number sold				
USA/worldwide	Not specified	Not specified	100	Not specified
Fiscal year	January to December	January to December	January to December	January to December
OTHER SPECIFICATIONS	Transport ventilator; outdoor usable (temperature range -18 to +50°C [-4 to +122°F], relative humidity 5-95%, ambient pressure 570-1,200 hPa). Vibration tested according to MIL STD 810 F (method 514.5). Airworthiness according to RTCA D-160D (section 7 and 8) IPX4.	All gas measurements under BTPS conditions; single rotary knob parameter adjustment; options and upgrades via software and/or hardware; open breathing system in all modes and patient ranges; low pressure oxygen inlet (LPO). Meets requirements of IEC 60601.	None specified.	Display tilts for better visibility when ventilator is pendant mounted and can be remotely mounted; automatic suction procedure, NIV capable; automatic patient detection; airway resistance compensation.
LAST UPDATED	April 2006	April 2006	April 2006	April 2006
Supplier Footnotes				
Model Footnotes				
Data Footnotes				¹ Bilevel VG not currently available in Canada and USA.

Product Comparison Chart

MODEL	GE HEALTHCARE (DATEX-OHMEDA) Centiva/5 (6.5" Screen)	GE HEALTHCARE (DATEX-OHMEDA) Engström Carestation	HAMILTON GALILEO GOLD	HAMILTON RAPHAEL COLOR
WHERE MARKETED	Worldwide ¹	Worldwide	Worldwide	Worldwide
FDA CLEARANCE	Yes	Yes	Yes	Yes
CE MARK (MDD)	Yes	Yes	Yes	Yes
PATIENT TYPE	Adult to pediatric, optional neonatal	Adult to infant	Adult, pediatric, neonatal	Adult, pediatric
CONTROLS				
Tidal volume, mL	20-2,000	20-2,000	10-2,000	50-2,000
Inspiratory flow, L/min	2-99.6, 180 maximum peak flow	2-160, 200 maximum peak flow	1-180	0-180
Inspiratory pressure, cm H ₂ O	1-59	1-98	0-100	0-50 above PEEP/CPAP
Respiratory rate, breaths/min	4-100	3-120 for control modes, 1-60 for support modes	0.5-120	0-99
Inspiratory time, sec	0.06-13.6, set via rate and I:E ratio	0.25-15	0.1-10	0.1-9.6
Expiratory time, sec	0.2-13.6, set via rate and I:E ratio	0.25-59.75	0.2-59.9	0.2-59.9
IE ratio	1:9 to 4:1, 9:1 in bilevel	1:9 to 4:1, 9:1 in bilevel	1:9 to 4:1, 150:1 in DuoPAP mode	9.9:1 to 1:9.9, 150:1 in DuoPAP mode
Inspiratory hold/plateau	Yes	Yes	0-70% cycle time	0-70% cycle time
Expiratory hold	No	Yes	10 sec maximum	NA
FiO ₂ , %	21-100	21-100	21-100	21-100
Manual breath	No	Yes	Yes	Yes
PEEP/CPAP, cm H ₂ O	Off, 2-35	Off, 1-50	0-50	0-35
Pressure support, cm H ₂ O	0-59	0-60	0-100	0-50 above PEEP/CPAP
Nebulizer	Pneumatic, synchronized, compensated	Built-in Aeroneb Pro Nebulizer System	Yes	Yes
Trigger mechanism	Flow	Pressure, flow	Pressure, flow	Flow
Bias/base flow range, L/min	3-30	2-10	1-30, automatic	1-10, automatic
Pressure slope/ramp adjustment	Rise time adjustment for inspiratory pressure and pressure support	Rise time adjustment for pressure, flow, and pressure support	25-200 msec	50-200 msec
Sigh	No	No	Yes	Yes
100% O ₂	Yes, suction maneuver	Yes, suction maneuver	Yes	Yes
Others	None specified	None specified	% tube resistance compensation, tube type/size	% tube resistance compensation, tube type/size

This is the first of four pages covering the above model(s). These specifications continue onto the next three pages.

Product Comparison Chart

MODEL	GE HEALTHCARE (DATEX-OHMEDA) Centiva/5 (6.5" Screen)	GE HEALTHCARE (DATEX-OHMEDA) Engström Carestation	HAMILTON GALILEO GOLD	HAMILTON RAPHAEL COLOR
OPERATING MODES				
Assist/control				
Volume breaths	Yes	Yes	Yes	Yes
Pressure breaths	Bilevel	Yes	Yes	Yes
SIMV				
Volume breaths	Yes	Yes	Yes	Yes
Pressure breaths	Bilevel	Yes	Yes	Yes
Pressure support	Yes	Yes	Yes	Yes
Spontaneous/CPAP				
Pressure support	Yes	Yes	Yes	Yes
Apnea-backup vent	Yes	Yes	Yes	Yes
Combination modes	Yes	Yes	Yes	Yes
Responsive valve	Not specified	Not specified	Yes	Yes
Bilevel/APRV	Yes	Yes	Yes	Yes
Others	Bilevel	Bilevel, pressure control with Volume Guarantee	AVtS, DuoPAP+, DuoPAP, NIV, APV, CMV/IMV	APRV, DuoPAP, DuoPAP+, NIV
MONITORED PARAMETERS				
Pressure				
PIP	Yes	Yes	Yes	Yes
MAP	Yes	Yes	Yes	Yes
PEEP	Yes	Yes	Yes	Yes
Volume				
Tidal	Yes	Yes	Yes	Yes
Minute	Yes	Yes	Yes	Yes
Spontaneous minute	Yes	Yes	Yes	Yes
FiO2	Yes	Yes	Yes	Yes
Respiratory rate	Yes	Yes	Yes	Yes
Inspiratory time	No	Yes	Yes	Yes
Expiratory time	No	Yes	Yes	Yes
IE	No	Yes	Yes	Yes
Others	Compliance, resistance, Pmin	Auxiliary pressure, patient flow, CO2, compliance, resistance, RQ, VO2, VCO2, RSBI, energy expenditure, 14-day trending, alarm logs	Static compliance, inspiratory/expiratory resistance, inspiratory/expiratory time constant, patient trigger, PO.1, RSBI (f/VT), WOB, pressure-time product, air trapping, auxiliary pressure, minimum pressure, Vleak, % O2	Inspiratory peak flow, compliance, expiratory resistance
PATIENT ALARMS				
FiO2	Yes	Yes	Yes	Yes
Low minute volume	Yes	Yes	Yes	Yes
Low inspiratory pressure	Yes	Yes	Yes	Not specified
High pressure	Yes	Yes	Yes	Yes
Loss of PEEP	Yes, indirectly	Yes	Yes	Yes
Apnea	Yes	Yes	Yes	Yes
High continuous pressure/occlusion	Yes	Yes	Yes	Yes
Inverse IE	Visual	Ventilation limit indicator	Text message	Text message
High respiratory rate	Yes	Yes	Yes	Yes
High minute volume	Yes	Yes	Yes	Yes
High PEEP	No	Yes	Not specified	Not specified
Breathing circuit disconnect	Yes	Yes	Yes	Yes
Others	Low respiratory rate	Low respiratory rate, high/low tidal volume, high intrinsic PEEP, ETCO2, ETO2, ventilator soft limit indicators	Low respiratory rate, low tidal volume, exhalation obstruction, maximum leak compensation, pressure limitation, check Ti, check %VE	Pressure limitation, high pressure during sigh, exhalation obstructed, disconnection

These specifications
continue onto the next two
pages

Product Comparison Chart

MODEL	GE HEALTHCARE (DATEX-OHMEDA) Centiva/5 (6.5" Screen)	GE HEALTHCARE (DATEX-OHMEDA) Engström Carestation	HAMILTON GALILEO GOLD	HAMILTON RAPHAEL COLOR
EQUIPMENT ALARMS				
Gas supply failure	Yes	Yes	Yes	Yes
Power failure	Yes	Yes	Yes	Yes
Vent inoperative	Yes	Yes	Yes	Yes
Low battery	Yes	Yes	Yes	Yes
Self-diagnostics	Yes	Yes	Yes	Yes
Others	Flow sensor, circuit leak, patient detected in standby, patient disconnect	Flow sensor, circuit leak, patient connect, patient disconnect, occlusion	Technical faults	Obstruction, maximum leak
INTERFACING				
Output ports	RS232	RS232, RS485, RS422, 2 USB ports, Ethernet, PC Flash card slot	RS232C	RS232C
Remote alarm/display	Yes, nurse call	No	Yes	Yes
Analog output	No	Yes	Yes	Yes
Report generation				
Display	Via N-DISVENT	Via N-DISVENT or Unity link	Optional (LEONARDO)	Optional (LEONARDO)
Hard copy	Via N-DISVENT	Via N-DISVENT or Unity link	No	No
Archival disk	No	Via N-DISVENT, Unity link, or PC Flash card slots	Event log-on flash drive	No
Network	No	Yes	Not specified	Not specified
DISPLAY TYPES	16.5 cm (6.5") diagonal color LCD	Color LCD 30.5 cm (12")	Color screen, LCD, TFT	Color screen, LCD, TFT
DATA DISPLAYED	Real-time pressure and flow waveforms, measured numerics, settings, status	Real-time graphics, numbers, 6 waveforms, 3 loops, trends, minitrends, take snapshot, split screens	Real-time graphics, numerics, waveforms, loops, trends	Real-time graphics, numerics, loops, trends
PNEUMATIC POWER				
Compressor	Optional	Optional	Optional	Optional
Compressed gases	O2, air	O2, air	O2, air; optional NO	O2, air
Pressure ranges	40-80 psi, 280-600 kPa	35-94 psi, 240-641 kPa	29-86 psi	29-86 psi
Turbine/piston	Not specified	Not specified	Not specified	Not specified
LINE POWER, VAC	98-132, 207-253; 50/60 Hz	85-132, 187-264; 47/63 Hz	100-240	100-240
Current, amps	Not specified	<1.7 (120 VAC), <0.87 (230 VAC)	2.3 (100 VAC), 1 (240 VAC)	0.7 (120 VAC)
Watts	<50	<200	230 maximum	40 VA
INTERNAL BATTERY	Yes	Yes	Yes	Yes
Type (number)	Sealed lead-acid (2)	Sealed lead-acid (2)	Sealed lead-acid (1)	Lead-acid (1)
Operating time, hr	0.5	0.5-2	1 minimum	1 minimum
Rechargeable	Yes	Yes	Yes	Yes
Recharging time, hr	12 (full discharge)	8 (full discharge)	6	6
EXTERNAL BATTERY	Yes	Not specified	Not specified	Not specified
Type (number)	Sealed lead-acid (2) in special battery pack	Not specified	Not specified	Not specified
Operating time, hr	1	Not specified	Not specified	Not specified
Rechargeable	Yes	Not specified	Not specified	Not specified
Recharging time, hr	10	Not specified	Not specified	Not specified
H x W x D, cm (in)	21 x 26 x 29 (8.3 x 10.2 x 11.4) CMS, 25 x 24 (9.8 x 9.4) footprint	123 x 38 x 36 (48.4 x 15 x 14) with trolley and display down; 144 x 38 x 36 (56.8 x 15 x 14) with trolley and display up	44 x 62 x 154 (17.3 x 24.5 x 61) with standard trolley	23.1 x 53 x 35 (9.1 x 20.9 x 13.8) without trolley
WEIGHT, kg (lb)	13 (28.7)	29 (64), 66 (145) with trolley	48 (105.8)	16.8 (37) without trolley

This is the third of four
pages covering the above
model(s). These
specifications continue
onto the next page.

Product Comparison Chart

MODEL	GE HEALTHCARE (DATEX-OHMEDA) Centiva/5 (6.5" Screen)	GE HEALTHCARE (DATEX-OHMEDA) Engström Carestation	HAMILTON GALILEO GOLD	HAMILTON RAPHAEL COLOR
PURCHASE INFORMATION				
List price	Not specified	Not specified	\$29,820	\$19,760
Warranty Service	1 year, parts and labor	1 year, parts and labor	1 year, parts and labor	1 year, parts and labor
Factory/on-site Training	Yes/yes	Yes/yes	Yes/yes	Yes/yes
3rd-party service	Yes	Yes	Yes	Yes
Parts availability	Yes	Yes	Yes	Yes
Delivery time, ARO	30-45 days	30-45 days	45-60 days	45-60 days
Year first sold	2002	2004	2005	2003
USA/worldwide	Not specified	Not specified	Not specified/1,268	Not specified/488
Fiscal year	January to December	January to December	December to November	December to November
OTHER SPECIFICATIONS	Display tilts for better visibility when ventilator is pendant mounted and can be remotely mounted; automatic suction procedure, NIV capable; automatic patient detection; airway resistance compensation.	Spontaneous Breathing Trial (SBT); nondepleting paramagnetic O2 sensor; user-customizable default settings; escalating high priority alarm; NIV capable; automatic patient detection; take snapshot; airway resistance compensation; programmable mode families; optional module bay provides enhanced respiratory monitoring.	Oxygen analyzer; smart apnea backup; proximal airway monitoring; TRC-automatic tube compensation; P/V tool; automated compliance curve with both inspiratory and expiratory limbs; ventilation analyzer window. Meets requirements of ANSI Z79.10-1979/ Z797-1976, CSA, DIN 13254, GLEM, IEC 60601 and 62D/60601, ISO 121/5359/5369/7767, JMMI, NFC 74-350, SEV, SETI, ORKI, VTT and TUV.	Meets requirements of ANSI Z79.10-1979/ Z797-1976, CSA, DIN 13254, GLEM, IEC 60601 and 62D/60601, ISO 121/5359/5369/7767, JMMI, NFC 74-350, SEV, and TUV.
LAST UPDATED	April 2006	April 2006	April 2006	April 2006
Supplier Footnotes				
Model Footnotes				
Data Footnotes	¹ Bilevel VG not currently available in Canada and USA.			

Product Comparison Chart

MODEL	IMPACT Uni-Vent 754	INTERMED INTER 5 PLUS/GMX	INTERMED INTER PLUS VAPS/GMX	KIMURA KV-3N
WHERE MARKETED	Worldwide	Worldwide	Worldwide	Asia, Middle East, South America, others
FDA CLEARANCE	Yes	No	No	No
CE MARK (MDD)	Yes	Yes	Yes	No
PATIENT TYPE	Not specified	Adult, pediatric, neonatal	Adult, pediatric, neonatal	Not specified
CONTROLS				
Tidal volume, mL	0-1,000	10-2,000	10-2,000	30-3,500
Inspiratory flow, L/min	0-60	2-120 (controlled), 0-150 (demand)	2-120 (controlled), 0-150 (demand)	10-70
Inspiratory pressure, cm H ₂ O	0-100	5-80	5-80	5-100
Respiratory rate, breaths/min	1-150	0-180	0-180	2-60 in 24 steps
Inspiratory time, sec	0.1-3	0.1-15	0.1-15	0.2-3 in 24 steps
Expiratory time, sec	NA	0.1 minimum (indirectly adjustable)	0.1 minimum (indirectly adjustable)	Not specified
IE ratio	1:1 to 1:600	1:99 to 9.9:1 (indirectly adjustable)	1:99 to 9.9:1 (indirectly adjustable)	1:0.5 to 1:149
Inspiratory hold/plateau	100% inspiratory time	Manual (15 sec maximum), 50% inspiratory time	Manual (15 sec maximum), 50% inspiratory time	Not specified
Expiratory hold	Not specified	Manual (15 sec maximum)	Manual (15 sec maximum)	Not specified
FiO ₂ , %	21-100	21-100	21-100	21-100
Manual breath	Yes	Yes (controlled cycle, started by operator)	Yes (controlled cycle, started by operator)	Not specified
PEEP/CPAP, cm H ₂ O	0-20	0-50	0-50	0-20
Pressure support, cm H ₂ O	Not specified	0-80	0-80	Not specified
Nebulizer	No	Yes (6 L/min, synchronized at inspiratory phase)	Yes (6 L/min, synchronized at inspiratory phase)	Yes
Trigger mechanism	Pressure	Pressure, flow	Pressure, flow	Not specified
Bias/base flow range, L/min	NA	Not specified	Not specified	0-40
Pressure slope/ramp adjustment	Yes	Automatic	Automatic	Not specified
Sigh	Yes	Yes (1.5 x tidal volume, each 100 cycles or 7 min)	Yes (1.5 x tidal volume, each 100 cycles or 7 min)	Yes
100% O ₂	Not specified	No	No	Not specified
Others	None specified	Inspiratory sensitivity by pressure (0.5-10 cm H ₂ O), inspiratory sensitivity by flow (0.2-15 L/min), TGI (6 L/min synchronized at expiratory phase), apnea interval (4-30 sec)	Inspiratory sensitivity by pressure (0.5-10 cm H ₂ O), inspiratory sensitivity by flow (0.2-15 L/min), TGI (6 L/min synchronized at expiratory phase), apnea interval (4-30 sec)	None specified

This is the first of four pages covering the above model(s). These specifications continue onto the next three pages.

Product Comparison Chart

MODEL	IMPACT Uni-Vent 754	INTERMED INTER 5 PLUS/GMX	INTERMED INTER PLUS VAPS/GMX	KIMURA KV-3N
OPERATING MODES				
Assist/control				
Volume breaths	Yes	Yes	Yes	No
Pressure breaths	No	Yes	Yes	No
SIMV				
Volume breaths	Yes	Yes	Yes	Yes
Pressure breaths	Yes	Yes	Yes	Yes
Pressure support	Not specified	Yes	Yes	Not specified
Spontaneous/CPAP				
Pressure support	No	Yes	Yes	No
Apnea-backup vent	Yes	Yes	Yes	No
Combination modes	Not specified	Not specified	Not specified	Not specified
Responsive valve	Not specified	Yes	Yes	Not specified
Bilevel/APRV	Not specified	Yes	Yes	Not specified
Others	None specified	Constant and decelerating wave flow, time cycle (A/C, SIMV), VAPS (A/C, SIMV)	Constant and decelerating wave flow, time cycle (A/C, SIMV), VAPS (A/C, SIMV)	None specified
MONITORED PARAMETERS				
Pressure				
PIP	Yes	Yes	Yes	Yes
MAP	Not specified	Yes	Yes	Yes
PEEP	Yes	Yes	Yes	Yes
Volume				
Tidal	Yes	Yes	Yes	Not specified
Minute	Yes	Yes	Yes	Not specified
Spontaneous minute	Not specified	Not specified	Not specified	Not specified
FiO2	Yes	Yes	Yes	Yes
Respiratory rate	Yes	Yes	Yes	Not specified
Inspiratory time	Yes	Yes	Yes	Not specified
Expiratory time	Yes	No	No	Not specified
IE	Yes	Yes	Yes	Not specified
Others	None specified	Tidal Volume (inspiratory and expiratory), plateau pressure, peak flow (inspiratory and expiratory), patient resistance (inspiratory and expiratory), patient compliance (static and dynamic), weaning index (RSBI), negative inspiratory pressure, work of breathing, graphics (PxT, FxT, VxT, PxV, FxV), trends (PIP, PEEP, rate, tidal volume, resistance, compliance)	Tidal Volume (inspiratory and expiratory), plateau pressure, peak flow (inspiratory and expiratory), patient resistance (inspiratory and expiratory), patient compliance (static and dynamic), weaning index (RSBI), negative inspiratory pressure, work of breathing, graphics (PxT, FxT, VxT, PxV, FxV), trends (PIP, PEEP, rate, tidal volume, resistance, compliance)	None specified
PATIENT ALARMS				
FiO2	Yes	Yes	Yes	Not specified
Low minute volume	Yes	Yes	Yes	Not specified
Low inspiratory pressure	Yes	Yes	Yes	Not specified
High pressure	Yes	Yes	Yes	Not specified
Loss of PEEP	Yes	No	No	Not specified
Apnea	Yes	Yes	Yes	Not specified
High continuous pressure/occlusion	Yes	Not specified	Not specified	Not specified
Inverse IE	Yes	No	No	Not specified
High respiratory rate	Yes	No	No	Not specified
High minute volume	Yes	No	No	Not specified
High PEEP	Not specified	Yes	Yes	Not specified
Breathing circuit disconnect	Not specified	Not specified	Not specified	Not specified
Others	None specified	None specified	None specified	None specified

These specifications continue onto the next two pages.

Product Comparison Chart

MODEL	IMPACT Uni-Vent 754	INTERMED INTER 5 PLUS/GMX	INTERMED INTER PLUS VAPS/GMX	KIMURA KV-3N
EQUIPMENT ALARMS				
Gas supply failure	Yes	Yes	Yes	Yes
Power failure	Yes	Yes	Yes	Yes
Vent inoperative	Yes	Yes	Yes	Not specified
Low battery	Yes	Yes	Yes	Not specified
Self-diagnostics	Sensor failure, CPU	Not specified	Not specified	Not specified
Others	Low battery, TV, valve leak	None specified	None specified	Electrical failure
INTERFACING				
Output ports	RS232	RS232	RS232	Not specified
Remote alarm/display	No	No	No	Not specified
Analog output	No	Not specified	Not specified	Not specified
Report generation				
Display	No	Not specified	Not specified	Not specified
Hard copy	No	Not specified	Not specified	Not specified
Archival disk	No	Not specified	Not specified	Not specified
Network	No	Not specified	Not specified	Not specified
DISPLAY TYPES	LED, LCD	7-segment numeric, LCD, 320 x 240 pixels, mono or color	7-segment numeric, LCD, 320 x 240 pixels, mono or color	Digital LEDs
DATA DISPLAYED	Numbers, graphics, waveforms, alarms	Numbers, text, graphics, loops, trends	Numbers, text, graphics, loops, trends	Not specified
PNEUMATIC POWER				
Compressor	Yes	Optional INTER 3500	Optional INTER 3500	Optional
Compressed gases	O ₂	O ₂ , air	O ₂ , air	O ₂ , air
Pressure ranges	40-80 psi	2.5-5 kg/cm ²	2.5-5 kg/cm ²	3.5-4.5 kg/cm ²
Turbine/piston	Not specified	No	No	Not specified
LINE POWER, VAC	95-265 autosensing	100-240	100-240	100/110/120/220/240, 50/60 Hz
Current, amps	5	0.50-0.22	0.50-0.22	Not specified
Watts	60	50	50	Not specified
INTERNAL BATTERY				
Type (number)	Yes	Yes	Yes	Not specified
	Sealed lead-acid (not specified)	Lead-acid 12 V/2.2 Ah (1)	Lead-acid 12 V/2.2 Ah (1)	Not specified
Operating time, hr	Not specified	2	2	Not specified
Rechargeable	Not specified	Yes	Yes	Not specified
Recharging time, hr	Not specified	8	8	Not specified
EXTERNAL BATTERY				
Type (number)	Not specified	Not specified	Not specified	Not specified
Operating time, hr	Not specified	Not specified	Not specified	Not specified
Rechargeable	Not specified	Not specified	Not specified	Not specified
Recharging time, hr	Not specified	Not specified	Not specified	Not specified
H x W x D, cm (in)	29.2 x 23.5 x 12.4 (11.5 x 9.3 x 4.9)	16 x 29 x 31.5 (6.3 x 11.4 x 12.4)	16 x 29 x 31.5 (6.3 x 11.4 x 12.4)	24.5 x 37 x 32 (9.6 x 14.6 x 12.6)
WEIGHT, kg (lb)	5.8 (12.8)	16 (35.3)	16 (35.3)	15 (33.1)

This is the third of four pages covering the above model(s). These specifications continue onto the next page.

Product Comparison Chart

MODEL	IMPACT Uni-Vent 754	INTERMED INTER 5 PLUS/GMX	INTERMED INTER PLUS VAPS/GMX	KIMURA KV-3N
PURCHASE INFORMATION				
List price	\$8,495	\$9,960-14,172	\$11,777-15,989	Not specified
Warranty	1 year	2 years	2 years	Not specified
Service				
Factory/on-site	No	Yes	Yes	Not specified
Training	Yes	Yes	Yes	Not specified
3rd-party service	No	Yes	Yes	Not specified
Parts availability	Yes	Yes	Yes	Not specified
Delivery time, ARO	2 weeks	30 days	30 days	Not specified
Year first sold	1997	2003	2002	Not specified
Number sold				
USA/worldwide	Not specified	>1,400 worldwide	>600 worldwide	NA/not specified
Fiscal year	January to December	Not specified	Not specified	Not specified
OTHER SPECIFICATIONS	Includes built-in air/oxygen mixer and compressor; altitude and PEEP compensation; automatic backlighting of LCD; pressure waveform; internal backup ventilator.	Complies with IEC 60601-1, IEC 60601-1-2, IEC 60601-2-12, and EN 794-1.	Complies with IEC 60601-1, IEC 60601-1-2, IEC 60601-2-12, and EN 794-1.	O2 concentration monitor; optional air compressor.
LAST UPDATED	April 2006	April 2006	April 2006	June 2002
Supplier Footnotes				
Model Footnotes				
Data Footnotes				

Product Comparison Chart

MODEL	KIMURA KV-5	MAQUET Servo-i (Adult : Infant)	MAQUET Servo-s (Adult : Pediatric)	NEWPORT E100M
WHERE MARKETED	Asia, Middle East, South America, others	Worldwide	Worldwide	Worldwide
FDA CLEARANCE	No	Yes	Yes	Yes
CE MARK (MDD)	No	Yes	Yes	Yes
PATIENT TYPE	Not specified	Adult, pediatric, neonatal	Adult, pediatric	Adult to neonatal
CONTROLS				
Tidal volume, mL	50-3,500	100-4,000 : 5-350	100-2,000	5-5,000
Inspiratory flow, L/min	10-70	0-200	0-200	1-100
Inspiratory pressure, cm H ₂ O	0-100	0-120	0-120	0-120
Respiratory rate, breaths/min	2-60	0-150	0-160	1-120
Inspiratory time, sec	0.3-3	0.1-5	0.1-5	0.1-3
Expiratory time, sec	Not specified	0-0.4 : 0-0.2	0-0.2, 0-0.4	By rate and inspiratory time
IE ratio	Not specified	1:10 to 4:1	1:10 to 4:1	1:99 to 4:1
Inspiratory hold/plateau	Plateau	0-30% respiratory cycle	0-30% respiratory cycle	Using pressure limit
Expiratory hold	Not specified	Yes	Yes	Not specified
FiO ₂ , %	21-100	21-100	21-100	21-100
Manual breath	Not specified	Yes	Yes	Yes
PEEP/CPAP, cm H ₂ O	0-30	0-50	0-50	0 to ≥25
Pressure support, cm H ₂ O	Not specified	0-100	0-100	Time-limited demand flow
Nebulizer	Yes	Optional	Optional	Yes
Trigger mechanism	Not specified	Pressure, flow	Pressure, flow	Pressure/autocontrol
Bias/base flow range, L/min	Not specified	0.5-2	0.5-2	Continuous 0-20+
Pressure slope/ramp adjustment	Not specified	Yes	Yes	Manual
Sigh	Yes	No	No	No
100% O ₂	Not specified	Yes	Yes	NA
Others	None specified	Inspiratory cycle off	Inspiratory cycle off	None specified

This is the first of four pages covering the above model(s). These specifications continue onto the next three pages.

Product Comparison Chart

MODEL	KIMURA KV-5	MAQUET Servo-i (Adult : Infant)	MAQUET Servo-s (Adult : Pediatric)	NEWPORT E100M
OPERATING MODES				
Assist/control				
Volume breaths	No	Yes	Yes	Yes
Pressure breaths	No	Yes	Yes	Yes
SIMV				
Volume breaths	Yes	Yes	Yes	Yes
Pressure breaths	Yes	Yes	Yes	Yes
Pressure support	Not specified	Yes	Yes	Time-limited demand flow
Spontaneous/CPAP				
Pressure support	Yes	Yes	Yes	Yes
Apnea-backup vent	Yes	Yes	Yes	Yes
Combination modes	Not specified	PRVC	PRVC	Not specified
Responsive valve	Not specified	Yes	Yes	Not specified
Bilevel/APRV	Not specified	BiVent	No	Not specified
Others	None specified	NIV, nCPAP, automode	NIV	Automatic trigger control
MONITORED PARAMETERS				
Pressure				
PIP	Yes	Yes	Yes	Yes
MAP	Yes	Yes	Yes	Yes
PEEP	Yes	Yes	Yes	Yes
Volume				
Tidal	Not specified	Yes	Yes	Set
Minute	Not specified	Yes	Yes	Optional
Spontaneous minute	Not specified	Yes	Yes	No
FiO2	Not specified	Yes	Yes	Set optional
Respiratory rate	Not specified	Yes	Yes	Yes
Inspiratory time	Not specified	Yes	Yes	Set
Expiratory time	Not specified	Yes	Yes	Set
IE	Yes	Yes	Yes	Yes
Others	Inverse I:E ratio	Open lung tool, 24 hr trends, systems event alarm, capnography option	Pressure and flow volume waveforms, trended data	None specified
PATIENT ALARMS				
FiO2	Not specified	Yes	Yes	Optional
Low minute volume	Not specified	Yes	Yes	Optional
Low inspiratory pressure	Not specified	Yes	Yes	Yes
High pressure	Not specified	Yes	Yes	Yes
Loss of PEEP	Not specified	Yes	Yes	No
Apnea	Not specified	Yes	Yes	Yes
High continuous pressure/occlusion	Not specified	Yes	Yes	Yes
Inverse IE	Not specified	Yes	Yes	Yes
High respiratory rate	Not specified	Yes	Yes	Optional
High minute volume	Not specified	Yes	Yes	Optional
High PEEP	Not specified	Yes	Yes	Yes
Breathing circuit disconnect	Not specified	Yes	Yes	Not specified
Others	None specified	Technical systems, patient flow overage, leakage	Technical systems	Optional high/low FiO2

This is the second of four pages covering the above model(s). These specifications continue onto the next two pages.

Product Comparison Chart

MODEL	KIMURA KV-5	MAQUET Servo-i (Adult : Infant)	MAQUET Servo-s (Adult : Pediatric)	NEWPORT E100M
EQUIPMENT ALARMS				
Gas supply failure	Yes	Yes	Yes	Yes
Power failure	Yes	Yes	Yes	Yes
Vent inoperative	Not specified	Yes	Yes	Yes
Low battery	Not specified	Yes	Yes	Yes
Self-diagnostics	Not specified	Yes	Yes	Yes
Others	Electrical failure	System continuous, pre-use	System continuous, pre-use	Remote alarm output, external alarm, silence cable, autoset alarms
INTERFACING				
Output ports	No	RS232C, nurse call	RS232C, nurse call	RS232
Remote alarm/display	NA	Yes	Yes	Yes
Analog output	NA	Yes	Yes	Yes
Report generation				
Display	NA	Optional	Optional	No
Hard copy	NA	Optional	Optional	No
Archival disk	NA	Optional	Optional	No
Network	NA	Yes	Yes	No
DISPLAY TYPES	Digital LEDs	Flat touchscreen	Flat touchscreen	LED, LCD
DATA DISPLAYED	Not specified	All monitored parameters	All monitored parameters	Numbers; optional graphic/waveform
PNEUMATIC POWER				
Compressor	Optional	Optional	Optional	Optional
Compressed gases	O2, air	O2, air	O2, air	O2, air
Pressure ranges	3.5-4.5 kg/cm ²	29-94 psi	29-94 psi	35-90 psig
Turbine/piston	Not specified	NA	NA	Not specified
LINE POWER, VAC	100/110/120/220/240, 50/60 Hz	100-120, 220-240; 50/60 Hz	100-120, 220-240; 50/60 Hz	100/120/220/240
Current, amps	Not specified	5	5	0.07-0.14
Watts	Not specified	40	40	Not specified
INTERNAL BATTERY				
Type (number)	Not specified	Yes	Yes	Internal
Operating time, hr	Not specified	Ni-MH (1-6 units)	12 V (not specified)	Sealed lead-acid (1)
Rechargeable	Not specified	0.5-3	1	6-8
Recharging time, hr	Not specified	Yes	Yes	Yes
		3 (full discharge)	6 (full discharge)	Not specified
EXTERNAL BATTERY				
Type (number)	Not specified	Optional	Optional	Not specified
Operating time, hr	Not specified	Optional	Optional	Not specified
Rechargeable	Not specified	Optional	Optional	Not specified
Recharging time, hr	Not specified	Optional	Optional	Not specified
H x W x D, cm (in)	39 x 44 x 29.5 (15.4 x 17.3 x 11.6)	41.5 x 30 x 20.5 (16.3 x 11.8 x 8.1) for patient unit	52 x 38 x 30 (20.5 x 15 x 11.8)	26.7 x 24.1 x 16.5 (10.5 x 9.5 x 6.5)
WEIGHT, kg (lb)	18 (39.7)	15 (33) for patient unit	18 (39.7)	8.2 (18)

This is the third of four pages covering the above model(s). These specifications continue onto the next page.

Product Comparison Chart

MODEL	KIMURA KV-5	MAQUET Servo-i (Adult : Infant)	MAQUET Servo-s (Adult : Pediatric)	NEWPORT E100M
PURCHASE INFORMATION				
List price	Not specified	Not specified	Not specified	Not specified
Warranty	1 year	1 year	1 year	3 years
Service				
Factory/on-site	Not specified	Yes/yes	Yes/yes	Yes/yes
Training	Not specified	Yes	Yes	Yes
3rd-party service	Not specified	Yes (under contract)	Yes (under contract)	Yes
Parts availability	Not specified	Yes	Yes	Yes
Delivery time, ARO	Not specified	45 days	45 days	30-60 days
Year first sold	Not specified	2001	2004	1997
Number sold				
USA/worldwide	NA/not specified	Not specified	Not specified	Not specified
Fiscal year	Not specified	January to December	January to December	January to December
OTHER SPECIFICATIONS	O2 concentration monitor and emergency backup unit; optional air compressor.	Transport capability. Meets the requirements of IEC 606011 and ISO 10651-1.	Transport capability. Meets the requirements of IEC 606011-2.	Built-in battery; automatic trigger control and alarm settings; time-limited demand flow allows delivery of mandatory flow to assist spontaneous patient breaths; for use on infants through adults; optional O2 and graphics monitor. Designed to meet requirements of CB, CSA, IEC, JIS, NRTL/C, and TUV.
LAST UPDATED	June 2002	April 2006	April 2006	April 2006
Supplier Footnotes				
Model Footnotes				
Data Footnotes				

Product Comparison Chart

MODEL	NEWPORT	NEWPORT	NEWPORT	PULMONETIC SYSTEMS/VIASYS HEALTHCARE LTV 900 : LTV 950 : LTV 1000
	E150 Breeze	e360	e500 Wave	
WHERE MARKETED	Worldwide	Worldwide	Worldwide	Worldwide
FDA CLEARANCE	Yes	Yes	Yes	Yes
CE MARK (MDD)	Yes	Yes	Yes	Yes
PATIENT TYPE	Adult to neonatal	Adult, pediatric, infant	Adult, pediatric, infant	Not specified
CONTROLS				
Tidal volume, mL	10-2,000	20-3,000 (5-3,000 in certain markets)	20-3,000 (5-3,000 in certain markets)	50-2,000
Inspiratory flow, L/min	1-120	1-180	1-180	>160
Inspiratory pressure, cm H ₂ O	0-120 VCV, 0-60 PCV	0-120 VCV, 0-80 PCV	0-120 VCV, 0-80 PCV	1-99
Respiratory rate, breaths/min	1-150	1-120 (1-150 in certain markets)	1-120 (1-150 in certain markets)	0-80
Inspiratory time, sec	0.1-3	0.1-5	0.1-5	0.3-9.9
Expiratory time, sec	3 + auto leak compensation up to 25 L/min	By rate and inspiratory time	By rate and inspiratory time	0.3-9.9
IE ratio	1:99 to 4:1, depends on rate and inspiratory time	1:99 to 4:1, depends on rate and inspiratory time	1:99 to 4:1, depends on rate and inspiratory time	1:4 to 4:1
Inspiratory hold/plateau	Not specified	Off; 0.1-2 sec, manual inspiratory hold for 5 sec (15 sec in certain markets) or automatic pause	Off; 0.5, 1, or 2 sec; manual inspiratory hold or automatic pause	NA : NA : 6 sec maximum
Expiratory hold	Not specified	Up to 20 sec	Yes	NA : NA : 4 sec maximum
FiO ₂ , %	21-100	21-100	21-100	NA : NA : 21-100
Manual breath	Yes	Yes	Yes	1 x current settings
PEEP/CPAP, cm H ₂ O	0-60	0-45	0-45	0-20
Pressure support, cm H ₂ O	Not specified	0-60	0-60	Off, 1-60
Nebulizer	Yes	Optional	Optional	Not specified
Trigger mechanism	Pressure	Pressure or flow	Pressure or flow	Flow
Bias/base flow range, L/min	0-40	3, auto leak compensation up to 25	3, auto leak compensation up to 25	10
Pressure slope/ramp adjustment	Manual	Automatic (except USA) and manual, pressure support has automatic (except USA) and manual expiratory threshold adjustment	Automatic (except USA) and manual; pressure support has automatic (except USA) and manual expiratory threshold adjustment	Yes
Sigh	Yes	Yes	Yes	No
100% O ₂	Not specified	Yes	Yes	No : No : Yes
Others	None specified	Open exhalation valve (BPRV) (in certain markets), NIV, flow wave, sigh, pause, compliance compensation	Open exhalation valve (BPRV), NIV, flow wave, sigh, pause, compliance compensation	None specified

This is the first of four pages covering the above model(s). These specifications continue onto the next three pages.

Product Comparison Chart

MODEL	NEWPORT	NEWPORT	NEWPORT	PULMONETIC SYSTEMS/VIASYS HEALTHCARE LTV 900 : LTV 950 : LTV 1000
	E150 Breeze	e360	e500 Wave	
OPERATING MODES				
Assist/control				
Volume breaths	Yes	Yes	Yes	Yes : Yes : Yes
Pressure breaths	Yes	Yes	Yes	No : Yes : Yes
SIMV				
Volume breaths	Yes	Yes	Yes	Yes
Pressure breaths	Yes	Yes	Yes	No : Yes : Yes
Pressure support	No	Yes	Yes	Yes
Spontaneous/CPAP				
Pressure support	Yes	Yes	Yes	Yes
Apnea-backup vent	No	Yes	Yes	Yes
Combination modes	Not specified	Yes	Yes	Not specified
Responsive valve	Not specified	Yes	Yes	Not specified
Bilevel/APRV	Not specified	Yes (in certain markets)	Yes	Not specified
Others	DuoFlow	Volume target pressure control and pressure support, NIV with any breath type/mode	Volume target pressure control and pressure support, NIV with any breath type/mode	None specified
MONITORED PARAMETERS				
Pressure				
PIP	Yes	Yes	Yes	Yes
MAP	Yes	Yes	Yes	Yes
PEEP	Yes	Yes	Yes	Yes
Volume				
Tidal	Set	Yes	Yes	Yes
Minute	Optional	Yes	Yes	Yes
Spontaneous minute	No	Yes	Yes	Yes
FiO2	Set optional	Yes	Yes (with autocalibration)	Set
Respiratory rate	Yes	Yes	Yes	Yes
Inspiratory time	Set	Yes	Yes	No
Expiratory time	Yes	Yes	Yes	No
IE	Yes	Yes	Yes	Yes
Others	None specified	Spontaneous rate, pulmonary mechanics, autoPEEP, plateau pressure; peak inspiratory and expiratory flow, waves, loops, and trends	Spontaneous rate, pulmonary mechanics, autoPEEP; plateau pressure peak for inspiratory and expiratory flow, waves, loops, and trends	None specified
PATIENT ALARMS				
FiO2	Optional	Yes	Yes	Yes
Low minute volume	Optional	Yes	Yes	Yes
Low inspiratory pressure	Yes	Yes	Yes	Yes
High pressure	Yes	Yes	Yes	Yes
Loss of PEEP	No	Yes	Yes	Not specified
Apnea	Yes	Yes	Yes	Yes
High continuous pressure/occlusion	Yes	Yes	Yes	Not specified
Inverse IE	Yes	Yes	Yes	Not specified
High respiratory rate	Optional	Yes	Yes	Yes
High minute volume	Optional	Yes	Yes	Not specified
High PEEP	No	Yes	Yes	Yes
Breathing circuit disconnect	Not specified	Yes	Not specified	Yes
Others	Optional high/low FiO2	None specified	O2 analyzer with auto calibration every 8 hr, operator alarms	None specified

This is the second of four pages covering the above model(s). These specifications continue onto the next two pages.

Product Comparison Chart

MODEL	NEWPORT	NEWPORT	NEWPORT	PULMONETIC SYSTEMS/VIASYS HEALTHCARE LTV 900 : LTV 950 : LTV 1000
	E150 Breeze	e360	e500 Wave	
EQUIPMENT ALARMS				
Gas supply failure	Yes	Yes	Yes	Yes
Power failure	Yes	Yes	Yes	Yes
Vent inoperative	Yes	Yes	Yes	Yes
Low battery	Yes	Yes	Yes	Yes
Self-diagnostics	Yes	Yes	Yes	Yes
Others	Remote alarm output	Remote alarm output, external alarm silence cable, flow sensor error	Remote alarm output, external alarm silence cable	High and low oxygen inlet pressure, external power low
INTERFACING				
Output ports	RS232	RS232, VGA, USB, remote alarm	RS232, printer	RS232; RJ11-4
Remote alarm/display	Yes	Yes	Yes	Yes
Analog output	Yes	Yes	Yes	No
Report generation				
Display	No	Yes	Yes	No
Hard copy	No	Yes	Yes	No
Archival disk	No	Yes	No	No
Network	No	Yes	Yes	No
DISPLAY TYPES				
	LED, LCD	LED, LCD	LED, LCD, vacuum fluorescent	LED, optional LCD
DATA DISPLAYED				
	Numbers; optional graphic/waveform	Numbers, graphic waveforms and loops, mechanics calculations	Numbers, graphic waveforms and loops, mechanics calculations	Numbers
PNEUMATIC POWER				
Compressor	Optional	Optional	Optional	Internal turbine
Compressed gases	O2, air	O2, air	O2, air (no O2 required if FiO2 @ 21%)	NA to power vent
Pressure ranges	35-90 psig	35-90 psig	35-90 psig	NA : NA : 40-80 psig
Turbine/piston	Not specified	No	No	Turbine
LINE POWER, VAC				
Current, amps	100/120/220/240 0.5-1	100/120/220/240 Not specified	100/120/220/240 0.07-0.14	90-240 3-5.5
Watts	45	Not specified	Not specified	36-66
INTERNAL BATTERY				
Type (number)	Internal Sealed lead-acid (1)	Yes Sealed lead-acid (1)	Internal Sealed lead-acid (1)	Yes Sealed lead-acid (1)
Operating time, hr	1	0.75	0.5	1
Rechargeable	Yes	Yes	Yes	Yes
Recharging time, hr	Not specified	Not specified	Not specified	Not specified
EXTERNAL BATTERY				
Type (number)	Not specified	Yes Not specified	Yes Not specified	Yes Lithium ion (1), sealed lead-acid (3)
Operating time, hr	Not specified	Not specified	Not specified	3-9
Rechargeable	Not specified	Not specified	Not specified	Yes
Recharging time, hr	Not specified	Not specified	Not specified	Not specified
H x W x D, cm (in)				
	25.4 x 33 x 27.9 (10 x 13 x 11)	Not specified	Varies; 3 modular components	8 x 25 x 30 (3.2 x 9.8 x 11.8)
WEIGHT, kg (lb)				
	13.6 (30)	Not specified	18.7 + 2.9 + 3.4 (41.3 + 6.4 + 7.5)	6.1 (13.4)

This is the third of four pages covering the above model(s). These specifications continue onto the next page.

Product Comparison Chart

MODEL	NEWPORT	NEWPORT	NEWPORT	PULMONETIC SYSTEMS/VIASYS HEALTHCARE LTV 900 : LTV 950 : LTV 1000
	E150 Breeze	e360	e500 Wave	
PURCHASE INFORMATION				
List price	Not specified	Not specified	Not specified	Not specified
Warranty	2 years	Not specified	1 year	1 year
Service				
Factory/on-site	Yes/yes	Yes/yes	Yes/yes	Yes/yes
Training	Yes	Yes	Yes	Yes
3rd-party service	Yes	Yes	Yes	Yes
Parts availability	Yes	Yes	Yes	Yes
Delivery time, ARO	30-60 days	30-60 days	30-60 days	3-5 days average
Year first sold	1988	2006	2001	1998
Number sold				
USA/worldwide	Not specified	Not specified	Not specified	Not specified
Fiscal year	January to December	January to December	January to December	Not specified
OTHER SPECIFICATIONS	1 hr emergency power; remote alarm; analog transducer output; optional air compressor; auxiliary flowmeter and AC outlets; O2 and graphics monitor available. Meets requirements of CSA 22.2.	None specified.	None specified.	Variable rise time and variable termination criteria for pressure- support and pressure- control breaths. Meets requirements of cETL, IEC 601-2-12, and ISO 13485.
LAST UPDATED	April 2006	April 2006	April 2006	April 2006
Supplier Footnotes				
Model Footnotes				
Data Footnotes				

Product Comparison Chart

MODEL	RESPIRONICS Esprit	SAIEM ELISEE	SIARE Siaretron 1000 ICU	SIARE Siaretron 1000 IPER
WHERE MARKETED	Worldwide	Europe	Worldwide, except USA	Worldwide, except USA
FDA CLEARANCE	Yes	No	No	No
CE MARK (MDD)	Yes	Yes	Yes	Yes
PATIENT TYPE	Adult, pediatric, neonatal	Not specified	Adult, pediatric, neonatal	Adult, pediatric, neonatal
CONTROLS				
Tidal volume, mL	50-2,500	50-2,500	5-3,000	5-3,000
Inspiratory flow, L/min	3-140 (compliance compensated, actual to 200)	5-180	0-120	0-120
Inspiratory pressure, cm H ₂ O	0-100	5-60	0-80	0-80
Respiratory rate, breaths/min	1-80 (adult, pediatric), 1-150 (neonatal)	1-80	1-150	1-150
Inspiratory time, sec	0.1-9.9 (0.1 resolution)	0.2-3	0.07-5	0.15-5
Expiratory time, sec	0.15-59.9	0.25-9	0.07-5	0.15-5
IE ratio	1:599 to 4:1	1:0.4 to 1:9.9	1:4 to 3:1	1:4 to 3:1
Inspiratory hold/plateau	0-2 sec (0.1 sec resolution)	0-5	0-50%	0-50%
Expiratory hold	Yes	0-12 sec	Yes	Yes
FiO ₂ , %	21-100	21-100	21-100	21-100
Manual breath	Yes	Yes	Not specified	Not specified
PEEP/CPAP, cm H ₂ O	0-35	0-25	0-20, 0-50 level 1, 0-25 level 2	0-20, 0-50 level 1, 0-25 level 2
Pressure support, cm H ₂ O	0-100 (relative to PEEP)	2-60	Yes	Yes
Nebulizer	Optional	Yes	Yes	Yes
Trigger mechanism	-20 to -0.1 cm H ₂ O (adult, pediatric), 0.5-20 L/min (adult, pediatric), 3.3-10.3 cm H ₂ O (neonatal), 0.1 cm H ₂ O resolution	Pressure, flow	Pressure, optional flow	Pressure, optional flow
Bias/base flow range, L/min	3.5-23 (adult, pediatric), 3.3-10.3 (neonatal)	Auto to 30	1-15 optional	1-15 optional
Pressure slope/ramp adjustment	0.1-0.9 sec (adult, pediatric), 0.1-0.5 (neonatal)	Yes	Yes	Yes
Sigh	No	Yes	Yes	Yes
100% O ₂	Yes (120 sec duration)	Yes	Yes	Yes
Others	None specified	None specified	None specified	None specified

This is the first of four pages covering the above model(s). These specifications continue onto the next three pages.

Product Comparison Chart

MODEL	RESPIRONICS Esprit	SAIME ELISEE	SIARE Siaretron 1000 ICU	SIARE Siaretron 1000 IPER
OPERATING MODES				
Assist/control				
Volume breaths	Yes	Yes	Yes	Yes
Pressure breaths	Yes	Yes	Yes	Yes
SIMV				
Volume breaths	Yes	Yes	Yes	Yes
Pressure breaths	Yes	Yes	No	No
Pressure support	Yes	Yes	Yes	Yes
Spontaneous/CPAP				
Pressure support	Yes	Yes	Yes	Yes
Apnea-backup vent	Yes	Yes	Yes	Yes
Combination modes	Not specified	Not specified	Not specified	Not specified
Responsive valve	Neonatal mode	Not specified	Not specified	Not specified
Bilevel/APRV	Not specified	Not specified	Yes	Yes
Others	Spontaneous/timed (NPPV), FlowTrak (VCV)	Pressure support, 5 preset modes	NIV	NIV
MONITORED PARAMETERS				
Pressure				
PIP	Yes	Yes	Yes	Yes
MAP	Yes	Yes	Yes	Yes
PEEP	Yes	Yes	Yes	Yes
Volume				
Tidal	Yes	Yes	Yes	No
Minute	Yes	Yes	Yes	No
Spontaneous minute	Yes	Yes	Yes	No
FiO2	Yes	Yes	Yes	No
Respiratory rate	Yes	Apnea, leaks	Yes	Yes
Inspiratory time	No	Yes	Yes	Yes
Expiratory time	No	Yes	Yes	Yes
IE	Yes	Yes	Yes	Yes
Others	% leak, spontaneous and total minute volume, spontaneous rate, measured end expiratory pressure, RSBI, TiTot, others	Events log	None specified	None specified
PATIENT ALARMS				
FiO2	High and low	Yes	Yes	No
Low minute volume	0-60 L/min	Yes	Yes	No
Low inspiratory pressure	3-105 cm H2O	Yes	Yes	Yes
High pressure	10-105 cm H2O	Yes	Yes	Yes
Loss of PEEP	0-35 cm H2O	Yes	No	No
Apnea	10-60 sec	Yes	Yes	Yes
High continuous pressure/occlusion	Yes	Yes	Yes	Yes
Inverse IE	Yes	Yes	No	No
High respiratory rate	0-150 bpm	Yes	Yes	No
High minute volume	0-60 L/min (adult, pediatric), 0-5 L/min (neonatal)	Yes	Yes	No
High PEEP	Yes	Yes	Yes	Yes
Breathing circuit disconnect	Not specified	Not specified	Yes	Yes
Others	None specified	None specified	None specified	None specified

This is the second of four pages covering the above model(s). These specifications continue onto the next two pages.

Product Comparison Chart

MODEL	RESPIRONICS Esprit	SAIME ELISEE	SIARE Siaretron 1000 ICU	SIARE Siaretron 1000 IPER
EQUIPMENT ALARMS				
Gas supply failure	Yes	Yes	Yes	Yes
Power failure	Yes	Yes	Yes	Yes
Vent inoperative	Yes	Yes	Yes	Yes
Low battery	Yes	Yes	Yes	Yes
Self-diagnostics	EST, SST	Yes	Sensor failure, wrong gas connection	Sensor failure, wrong gas connection
Others	120 sec alarm silence, reset	Exhale valve, low battery, temperature	Valve failure	Valve failure
INTERFACING				
Output ports	RS232	Bidirectional RS232	RS232	RS232
Remote alarm/display	Yes (nurse call)	Yes	No	No
Analog output	Yes	No	No	No
Report generation				
Display	Optional	15" color screen	Yes	Yes
Hard copy	Yes	No	No	No
Archival disk	No	No	No	No
Network	No	RS232	SIARELINK	SIARELINK
DISPLAY TYPES	26.4 cm (10.4") color touchscreen	EL touchscreen, TFT touchscreen	LED, LCD	LED, LCD
DATA DISPLAYED	Numbers, waveform, loops, respiratory mechanics, trended data, NICO interfaced data	Numbers, waveforms, loops, trends	Numbers	Numbers
PNEUMATIC POWER				
Compressor	Not needed	Internal microblower	Optional	No
Compressed gases	O2 (50 psi), air (blower)	O2	O2, air	O2, air
Pressure ranges	40-90 psi	3-6 bar, 45-90 psi	3.5 ±0.75 bar	3.5 ±0.75 bar
Turbine/piston	Blower	Not specified	No	No
LINE POWER, VAC	100-240	110-230, 50/60 Hz	100-240, 12 VDC	12 VDC
Current, amps	6 maximum	1-0.6	0.15	1.2
Watts	30	75 maximum	30	15
INTERNAL BATTERY	Yes	Yes	Yes	Yes
Type (number)	Lead-acid (not specified)	Lithium ion (not specified)	Lead-acid (1)	Ni-Cd (2)
Operating time, hr	30	3-4	2	6
Rechargeable	Yes	5	Yes	Yes
Recharging time, hr	~10	5	Not specified	Not specified
EXTERNAL BATTERY	Yes	Yes	Yes	No
Type (number)	Lead acid (not specified)	Lithium ion (not specified)	Lead-acid (1)	NA
Operating time, hr	2.5	6-8	10	NA
Rechargeable	Yes	5	Yes	NA
Recharging time, hr	~15	5	Not specified	NA
H x W x D, cm (in)	40 x 36 x 61 (16 x 14 x 24)	26 x 25 x 13 (10.2 x 9.8 x 5.1)	16 x 36 x 35 (6.2 x 14.2 x 13.8)	16 x 36 x 35 (6.2 x 14.2 x 13.8)
WEIGHT, kg (lb)	30 (66.2)	4.4 (9.7), ventilator and cart	4.9 (10.8)	4.9 (10.8)

This is the third of four pages covering the above model(s). These specifications continue onto the next page.

Product Comparison Chart

MODEL	RESPIRONICS Esprit	SAIME ELISEE	SIARE Siaretron 1000 ICU	SIARE Siaretron 1000 IPER
PURCHASE INFORMATION				
List price	Not specified	Not specified	Not specified	Not specified
Warranty	2 years	2 years	2 years	2 years
Service				
Factory/on-site	Yes/yes	Yes	Not specified	Not specified
Training	Yes, user configured	Yes	Not specified	Not specified
3rd-party service	Yes	No	Not specified	Not specified
Parts availability	Yes	Yes	Not specified	Not specified
Delivery time, ARO	2-4 weeks	Not specified	Not specified	Not specified
Year first sold	1999	2004	Not specified	Not specified
Number sold				
USA/worldwide	Not specified	Not specified	Not specified	Not specified
Fiscal year	July to June	July to June	January to December	January to December
OTHER SPECIFICATIONS	Meets requirements of CEI Class 1 Type B, EN 1281-1, IEC 60601-1, IEC 60601-1-1, IEC 60601-1-2, IEC 60601-2-12, and ISO 5356. CSA and CE mark shown.	None specified.	Meets requirements of CEI Class 1 Type B, EN 1281-1, IEC 60601-1, IEC 60601-1-1, IEC 60601-1-2, IEC 60601-2-12, and ISO 5356.	For use in hyperbaric chamber; autocompensated up to 60 mL H ₂ O. Meets the requirements of CEI Class 1 Type B, IEC 60601-1, IEC 60601-1-2, IEC 60601-2-12, and ISO 5369.
LAST UPDATED	April 2006	April 2006	April 2006	April 2006
Supplier Footnotes				
Model Footnotes				
Data Footnotes				

Product Comparison Chart

MODEL	SIARE Siaretron 3000 ICU	TAEMA eXtend	TAEMA Horus 4	TAEMA NEFTIS ICU
WHERE MARKETED	Worldwide, except USA	Worldwide, except USA	Worldwide, except USA	Worldwide, except USA
FDA CLEARANCE	No	No	No	No
CE MARK (MDD)	Yes	Yes	Yes	Yes
PATIENT TYPE	Adult, pediatric, neonatal	Adult to infant	Adult to infant	Adult to infant
CONTROLS				
Tidal volume, mL	5-3,000	20-2,000; 5,000 maximum	20-2,000; 5,000 maximum	40-2,000; 5,000 maximum
Inspiratory flow, L/min	0-120	1-200	1-200	1-200
Inspiratory pressure, cm H ₂ O	0-80	0-90	0-90	0-100
Respiratory rate, breaths/min	1-150	1-80	1-80	1-120
Inspiratory time, sec	0.07-5	0.2-5	0.2-5	0.2-10
Expiratory time, sec	0.07-5	20-90% cycle time	20-90% cycle time	20-90% cycle time
IE ratio	1:4 to 3:1	1:4 to 4:1	1:4 to 4:1	1:4 to 4:1
Inspiratory hold/plateau	0-50%	0-60% cycle time	0-60% cycle time	0-60% cycle, manual
Expiratory hold	Yes	Yes	Yes	Yes
FiO ₂ , %	21-100	21-100	21-100	21-100
Manual breath	Not specified	No	No	No
PEEP/CPAP, cm H ₂ O	0-20, 0-50 level 1, 0-25 level 2	0-40	0-25	0-50
Pressure support, cm H ₂ O	Yes	0-40	0-40	0-40
Nebulizer	Yes	Yes	Yes	Yes
Trigger mechanism	Pressure, optional flow	Pressure, flow	Pressure, flow	Pressure, flow
Bias/base flow range, L/min	1-15 optional	0-30, automatic	0-30, automatic	0-30, automatic
Pressure slope/ramp adjustment	Yes	50-150	50-150	50-200
Sigh	Yes	Yes	Yes	Yes
100% O ₂	Yes	No	No	Yes
Others	None specified	None specified	None specified	None specified

This is the first of four pages covering the above model(s). These specifications continue onto the next three pages.

Product Comparison Chart

MODEL	SIARE Siaretron 3000 ICU	TAEMA eXtend	TAEMA Horus 4	TAEMA NEFTIS icu
OPERATING MODES				
Assist/control				
Volume breaths	Yes	Yes	Yes	Yes
Pressure breaths	Yes	Yes	Yes	Yes
SIMV				
Volume breaths	Yes	Yes	Yes	Yes
Pressure breaths	No	Yes	Yes	Yes
Pressure support	Yes	Yes	Yes	Yes
Spontaneous/CPAP				
Pressure support	Yes	Yes	Yes	Yes
Apnea-backup vent	Yes	Yes	Yes	Yes
Combination modes	Not specified	Not specified	Not specified	Not specified
Responsive valve	Not specified	Not specified	Not specified	Not specified
Bilevel/APRV	Yes	Not specified	Not specified	Not specified
Others	NIV	PS-PEEP with minivent rate, inverse I:E, VAPS, PRVC, APRV, noninvasive, MRV	PS-PEEP with minivent rate, inverse I:E, VAPS, PRVC, APRV, noninvasive, MRV	PS-PEEP with minivent rate, inverse I:E, noninvasive
MONITORED PARAMETERS				
Pressure				
PIP	Yes	Yes	Yes	Yes
MAP	Yes	Yes	Yes	No
PEEP	Yes	Yes	Yes	Yes
Volume				
Tidal	Yes	Yes	Yes	Yes
Minute	Yes	Yes	Yes	Yes
Spontaneous minute	Yes	Yes	Yes	Yes
FiO2	Yes	Yes	Yes	Yes
Respiratory rate	Yes	Yes	Yes	Yes
Inspiratory time	Yes	Yes	Yes	No
Expiratory time	Yes	Yes	Yes	No
IE	Yes	Yes	Yes	No
Others	None specified	f/VT, PV curve with low inflation flow method, leaks index, Pplat, VTi, VTe	f/VT, PV curve with low inflation flow method, leaks index, Pplat, VTi, VTe	Pplat, VTi, VTe
PATIENT ALARMS				
FiO2	Yes	Yes	Yes	Yes
Low minute volume	Yes	Yes	Yes	Yes
Low inspiratory pressure	Yes	Yes	Yes	Yes
High pressure	Yes	Yes	Yes	Yes
Loss of PEEP	No	Yes	Yes	Yes
Apnea	Yes	Yes	Yes	Yes
High continuous pressure/occlusion	Yes	Yes	Yes	Yes
Inverse IE	No	Yes	Yes	No
High respiratory rate	Yes	Yes	Yes	Yes
High minute volume	Yes	Yes	Yes	Yes
High PEEP	Yes	Yes	Yes	Yes
Breathing circuit disconnect	Yes	Yes	Yes	Yes
Others	None specified	Gas disconnect, battery alarms	Gas disconnect	Gas disconnect, battery alarms

This is the second of four pages covering the above model(s). These specifications continue onto the next two pages.

Product Comparison Chart

MODEL	SIARE Siaretron 3000 ICU	TAEMA eXtend	TAEMA Horus 4	TAEMA NEFTIS icu
EQUIPMENT ALARMS				
Gas supply failure	Yes	Yes	Yes	Yes
Power failure	Yes	Yes	Yes	Yes
Vent inoperative	Yes	Yes	Yes	Yes
Low battery	Yes	Yes (optional for UPS)	Yes (optional for UPS)	Yes
Self-diagnostics	Sensor failure, wrong gas connection	Yes	Yes	Yes
Others	Valve failure	Messages, alarms, history	Messages, alarms, history	Messages, alarms, history
INTERFACING				
Output ports	RS232	RS232 (2)	RS232 (2)	RS232 (2)
Remote alarm/display	No	Yes	Yes	Yes
Analog output	No	No	No	No
Report generation				
Display	Yes	Yes	Optional, accessed through optional Horus MdV monitor	No
Hard copy	No	No	Optional, accessed through optional Horus MdV monitor	No
Archival disk	No	No	Optional, accessed through optional Horus MdV monitor	No
Network	SIARELINK	Vital signs monitor	Vital signs monitor	No
DISPLAY TYPES	12.7 cm (5") graphic display, LED, LCD	15" color touchscreen	LCD (2), optional VGA color touchscreen	Color touchscreen
DATA DISPLAYED	Numbers, waveform, loops, respiratory mechanics	Numbers, graphics, waveforms, trends, PV curve	Numbers, graphics, waveforms, trends	Numbers, graphics, waveforms, trends
PNEUMATIC POWER				
Compressor	Optional	Optional	Optional	Turbine
Compressed gases	O2, air	O2, air	O2, air	O2
Pressure ranges	3.5 ±0.75 bar	2.8-6 bar	2.8-6 bar	2.8-6 bar
Turbine/piston	No	No	No	Yes
LINE POWER, VAC	100-240, 12 VDC	10-240, 50/60 Hz	10-240, 50/60 Hz	10-240, 50/60 Hz
Current, amps	0.15	Not specified	0.4	Not specified
Watts	30	90 VA	75 VA	200
INTERNAL BATTERY	Yes	Yes	No	Yes
Type (number)	Lead-acid (1)	Sealed lead-acid (not specified)	NA	Ni-MH (not specified)
Operating time, hr	2	0.5	NA	1.5
Rechargeable	Yes	Yes	NA	Yes
Recharging time, hr	Not specified	8	NA	12
EXTERNAL BATTERY	Yes	Optional	Optional	Optional
Type (number)	Lead-acid (1)	Any (not specified)	Not specified	Any (not specified)
Operating time, hr	10	Not specified	0.5	Not specified
Rechargeable	Yes	Not specified	Yes	Not specified
Recharging time, hr	Not specified	Not specified	8	Not specified
H x W x D, cm (in)	32 x 36 x 35 (12.5 x 14.2 x 13.8)	50 x 45 x 28 (19.7 x 17.7 x 11)	20 x 43 x 44 (7.9 x 16.9 x 17.3)	50 x 30 x 26 (19.7 x 11.8 x 10.2)
WEIGHT, kg (lb)	5.2 (11.5)	27 (59.5)	14 (30)	17 (37.5)

This is the third of four pages covering the above model(s). These specifications continue onto the next page.

Product Comparison Chart

MODEL	SIARE Siaretron 3000 ICU	TAEMA eXtend	TAEMA Horus 4	TAEMA NEFTIS icu
PURCHASE INFORMATION				
List price	Not specified	Not specified	Not specified	Not specified
Warranty	2 years	1 year	1 year	1 year
Service				
Factory/on-site	Not specified	Yes/yes	Yes/yes	Yes/yes
Training	Not specified	Yes	Yes	Yes
3rd-party service	Not specified	No	No	No
Parts availability	Not specified	Yes	Yes	Yes
Delivery time, ARO	Not specified	Not specified	Not specified	Not specified
Year first sold	Not specified	2005	1997	2006
Number sold				
USA/worldwide	Not specified	Not specified	NA/>4,000	Not specified
Fiscal year	January to December	Not specified	Not specified	Not specified
OTHER SPECIFICATIONS	Meets requirements of CEI Class 1 Type B, EN 1281-1, IEC 60601-1, IEC 60601-1-1, IEC 60601-1-2, IEC 60601-2-12, and ISO 5356.	Compliance-compensated tubing; optional software upgrades.	Compliance-compensated tubing; optional software upgrades.	Compliance-compensated tubing; optional software upgrades.
LAST UPDATED	April 2006	April 2006	April 2006	April 2006
Supplier Footnotes				
Model Footnotes				
Data Footnotes				

Product Comparison Chart

MODEL	TECME	TYCO HEALTHCARE PURITAN BENNETT	TYCO HEALTHCARE PURITAN BENNETT	TYCO HEALTHCARE PURITAN BENNETT
	Neumovent Graph	740	760	840
WHERE MARKETED	Worldwide, except USA	Worldwide	Worldwide	Worldwide
FDA CLEARANCE	No	Yes	Yes	Yes
CE MARK (MDD)	Yes	Yes	Yes	Yes
PATIENT TYPE	Adult, pediatric, neonatal	Not specified	Not specified	Adult to neonate
CONTROLS				
Tidal volume, mL	10-2,500	40-2,000	40-2,000	5-2,500
Inspiratory flow, L/min	0-180	3-150	3-150	3-150
Inspiratory pressure, cm H ₂ O	120	5-80	5-80	5-90
Respiratory rate, breaths/min	1-150	1-70	1-70	1-150
Inspiratory time, sec	0.1-30	0.2-8	0.2-8	0.2-8
Expiratory time, sec	Not specified	>0.2	>0.2	>0.2
IE ratio	60:1 to 1:199	1:99 to 4:1	1:99 to 4:1	1:299 to 4:1
Inspiratory hold/plateau	0-2 sec	0-2 sec	0-2 sec	0-2
Expiratory hold	Not specified	Not specified	Not specified	0.5-3 sec automatic pause, <20 sec manual pause
FiO ₂ , %	21-100	21-100	21-100	21-100
Manual breath	Yes	Yes	Yes	Yes
PEEP/CPAP, cm H ₂ O	0-50	0-35	0-35	0-45
Pressure support, cm H ₂ O	0-70	Yes	Yes	0-70
Nebulizer	Yes	Ultrasonic	Ultrasonic	Ultrasonic
Trigger mechanism	Pressure, flow	Flow	Flow	Pressure, flow
Bias/base flow range, L/min	1-10	NA	NA	1.5 above flow sensor
Pressure slope/ramp adjustment	Yes	Yes	Yes	Yes
Sigh	Yes	No	No	No
100% O ₂	Yes	Not specified	Not specified	Yes
Others	None specified	None specified	None specified	None specified

This is the first of four pages covering the above model(s). These specifications continue onto the next three pages.

Product Comparison Chart

MODEL	TECME	TYCO HEALTHCARE PURITAN BENNETT 740	TYCO HEALTHCARE PURITAN BENNETT 760	TYCO HEALTHCARE PURITAN BENNETT 840
	Neumovent Graph			
OPERATING MODES				
Assist/control				
Volume breaths	Yes	No	Yes	Yes
Pressure breaths	Yes	No	Yes	Yes
SIMV				
Volume breaths	Yes	Yes	Yes	Yes
Pressure breaths	Yes	Yes	Yes	Yes
Pressure support	Yes	Not specified	Not specified	Not specified
Spontaneous/CPAP				
Pressure support	Yes	Yes	Yes	Yes
Apnea-backup vent	Yes	Yes	Yes	Yes
Combination modes	Yes	Not specified	Not specified	Yes, VV+
Responsive valve	Not specified	Not specified	Not specified	Yes, active exhalation valve
Bilevel/APRV	Yes	Not specified	Not specified	Yes
Others	TCPL, MMV, NIV	Occlusion cycling, disconnect	Occlusion cycling, disconnect	Occlusion cycling, disconnect; NIV and PAV+ outside of USA and FDA 510k submitted in USA
MONITORED PARAMETERS				
Pressure				
PIP	Yes	Yes	Yes	Yes
MAP	Yes	Yes	Yes	Yes
PEEP	Yes	Yes	Yes	Yes
Volume				
Tidal	Yes	Yes	Yes	Yes
Minute	Yes	Yes	Yes	Yes
Spontaneous minute	Yes	Yes	Yes	Yes
FiO2	Yes	Yes	Yes	Yes
Respiratory rate	Yes	Yes	Yes	Inspiratory time too long
Inspiratory time	Yes	Yes	Yes	Yes
Expiratory time	Yes	Yes	Yes	Yes
IE	Yes	Yes	Yes	Yes
Others	Lung mechanics, 24 hr trends, systems event alarm	None specified	E-sense	E-sense, rapid shallow breathing index, Ti/Ttot
PATIENT ALARMS				
FiO2	Yes	Yes, ±10% of setting	Yes, ±10% of setting	Yes
Low minute volume	Yes	0-50 L	0-50 L	Yes
Low inspiratory pressure	Yes	3-60 cm H2O	3-60 cm H2O	Yes
High pressure	Yes	10-90 cm H2O	10-90 cm H2O	Yes
Loss of PEEP	Yes	Not specified	Not specified	Yes
Apnea	Yes	10-60 sec	10-60 sec	Yes
High continuous pressure/occlusion	Yes	Defaults to HP limit	Defaults to HP limit	Yes
Inverse IE	Not specified	Not specified	Not specified	Yes
High respiratory rate	Yes	Yes	Yes	Yes
High minute volume	Yes	Low minute, 0-50 L	Low minute, 0-50 L	Yes
High PEEP	Yes	Not specified	Not specified	Yes
Breathing circuit disconnect	Yes	Yes	Yes	Yes
Others	High tidal volume, high and low exhaled minute volume	None specified	None specified	High exhaled tidal volume, low exhaled mandatory tidal volume, low exhaled spontaneous tidal volume, inspiration too long, high delivered oxygen, low delivered oxygen

This is the second of four pages covering the above model(s). These specifications continue onto the next two pages.

Product Comparison Chart

MODEL	TECME	TYCO HEALTHCARE PURITAN BENNETT	TYCO HEALTHCARE PURITAN BENNETT	TYCO HEALTHCARE PURITAN BENNETT
	Neumovent Graph	740	760	840
EQUIPMENT ALARMS				
Gas supply failure	Yes	Yes	Yes	Yes
Power failure	Yes	Yes	Yes	Yes
Vent inoperative	Yes	Yes	Yes	Yes
Low battery	Yes	Yes	Yes	Yes
Self-diagnostics	Yes	Electronics and pneumatics	Electronics and pneumatics	Electronics and pneumatics
Others	Technical failure, fan failure	Disconnect, occlusion	Disconnect, occlusion	Disconnect, occlusion, AC power loss, compressor inoperative, device alert, low battery, no air supply, no oxygen supply, procedure error, screen block
INTERFACING				
Output ports	RS232C	RS232 (2)	RS232 (2)	RS232 (2)
Remote alarm/display	No	Yes	Yes	Yes
Analog output	No	No	No	No
Report generation				
Display	Yes	Optional	Optional	Yes
Hard copy	Yes	Yes	Yes	Yes
Archival disk	Yes	Yes	Yes	Yes
Network	No	Yes	Yes	Yes
DISPLAY TYPES	Color LCD	LCD	LCD	Color LCD, touchscreen
DATA DISPLAYED	Numbers, graphics, waveforms, trends	Numbers, optional graphics	Numbers, optional graphics	Numbers, graphics, waveforms, alarm violations
PNEUMATIC POWER				
Compressor	Optional	No	No	Optional
Compressed gases	O2, air	O2	O2	O2, air
Pressure ranges	50-100 psi	40-90 psi	40-90 psi	35-100 psi
Turbine/piston	Not specified	Piston	Piston	Not specified
LINE POWER, VAC	95-240	100-240	100-240	100-240
Current, amps	0.3	2.9 (115 V)	2.9 (115 VAC)	4.5 (115 VAC)
Watts	30	290	290	1,000
INTERNAL BATTERY				
Type (number)	Yes	Yes	Yes	Yes
Operating time, hr	Sealed lead-acid (1)	Not specified	Not specified	Not specified
	2	2.5 under normal conditions	2.5 under normal conditions	Not specified
Rechargeable	Yes	Yes	Yes	Not specified
Recharging time, hr	8	Not specified	Not specified	Not specified
EXTERNAL BATTERY				
Type (number)	No	Yes	Yes	Yes
Operating time, hr	NA	Not specified	Not specified	Not specified
Rechargeable	NA	7 under normal conditions	7 under normal conditions	At least 0.5
Recharging time, hr	NA	Yes	Yes	Yes
	NA	Not specified	Not specified	8 maximum, automatic when connected to AC power
H x W x D, cm (in)	31 x 35 x 23 (12.2 x 13.8 x 9.1)	37.8 x 51.5 x 37 (14.9 x 20.3 x 14.6), 25.5 x 51.5 x 37 (49.4 x 20.3 x 14.6) with cart	37.8 x 51.5 x 37 (14.9 x 20.3 x 14.6), 125.5 x 51.5 x 37 (49.4 x 20.3 x 14.6) with cart	33 x 45.7 x 25.4 (13 x 18 x 10) breath-delivery unit, 46 x 39.4 x 17 (18.1 x 15.5 x 6.7) graphical user interface, 99.8 x 58.2 x 60.2 (39.3 x 22.9 x 23.7) cart
WEIGHT, kg (lb)	12.5 (27.6)	30 (66.2) ventilator, 18 (39.7) cart	30 (66.2) ventilator, 18 (39.7) cart	18.2 (40.1) breath-delivery unit, 5.7 (12.6) graphical user interface, 15.5 (34.2) cart

This is the third of four pages covering the above model(s). These specifications continue onto the next page.

Product Comparison Chart

MODEL	TECME	TYCO HEALTHCARE PURITAN BENNETT	TYCO HEALTHCARE PURITAN BENNETT	TYCO HEALTHCARE PURITAN BENNETT
	Neumovent Graph	740	760	840
PURCHASE INFORMATION				
List price	\$10,000-13,000	\$14,500	\$18,765	\$32,120
Warranty	1 year	2 years	2 years	2 years
Service				
Factory/on-site	Yes/yes	Yes/yes	Yes/yes	Yes/yes
Training	Yes	Yes	Yes	Yes
3rd-party service	Yes	Yes	Yes	Yes
Parts availability	Yes	Yes	Yes	Yes
Delivery time, ARO	30 days	45 days	45 days	45 days
Year first sold	1999	1997	1998	1998
Number sold				
USA/worldwide	NA/not specified	Not specified	Not specified	Not specified
Fiscal year	March to February	July to June	July to June	October to September
OTHER SPECIFICATIONS	Optional heater and humidifier; foreign-language menus; electronic watchdog. Meets requirements of IEC 60601-1 and IEC 60601-2-12.	Integral O2 sensor; 2-5 hr battery backup; optional 7 hr external battery backup. Meets requirements of CISPR 11, CSA, IEC 60601-1, IEC 60601-2, IEC 60601-2-12, and UL.	Exhalation sensitivity; rise time percentage; integral O2 sensor; 2-5 hr battery backup; optional 7 hr external battery backup. Meets requirements of CISPR 11, CSA, IEC 60601-1, IEC 60601-2, and UL.	Exhalation sensitivity; rise time percentage; integral O2 sensor; 30 min battery backup. Meets requirements of CISPR 11, CSA, IEC 60601-1, IEC 60601-2, IEC 60601-2-12, and UL.
LAST UPDATED	April 2006	April 2006	April 2006	April 2006
Supplier Footnotes				
Model Footnotes				
Data Footnotes				

Product Comparison Chart

MODEL	VERSAMED iVent 201	VIASYS HEALTHCARE AVEA	VIASYS HEALTHCARE Vela	VIASYS HEALTHCARE Vela +
WHERE MARKETING	Worldwide	Worldwide	Worldwide	Worldwide
FDA CLEARANCE	Yes	Yes	Yes	Yes
CE MARK (MDD)	Yes	Yes	Yes	Yes
PATIENT TYPE	Adult, pediatric	Adult, pediatric, neonatal	Adult, pediatric	Adult, pediatric
CONTROLS				
Tidal volume, mL	50-2,000	100-2,500	50-2,000	50-2,000
Inspiratory flow, L/min	1-120, spontaneous 180	0.4-30, 1-75, 3-150, 180 spontaneous maximum	10-140/180 spontaneous maximum	10-140/180 spontaneous maximum
Inspiratory pressure, cm H ₂ O	0-80	0-80, 0-90	0-100	0-100
Respiratory rate, breaths/min	1-80	1-150, 1-120	2-80	2-80
Inspiratory time, sec	0.3-3, adaptive time	0.15-3, 0.2-5	0.3-10	0.3-10
Expiratory time, sec	Not specified	Depends on rate	Depends on rate	Depends on rate
IE ratio	Adjustable	4:1 maximum inverse	4:1 maximum inverse	4:1 maximum inverse
Inspiratory hold/plateau	Yes	0-3	Off, 0.1-2 sec	Off, 0.1-2 sec
Expiratory hold	Yes	3/20 sec	Manual	Manual
FiO ₂ , %	21-100	21-100	21-100	21-100
Manual breath	Yes	Yes	Yes	Yes
PEEP/CPAP, cm H ₂ O	0-40	0-50	0-35	0-35
Pressure support, cm H ₂ O	0-60	0-80 neonate, 0-90 adult/pediatrics	Off, 1-60	Off, 1-60
Nebulizer	Yes	20 minutes	Off, 1 to 60 min	Off, 1 to 60 minutes
Trigger mechanism	Pressure, flow, dual	Pressure, flow	Flow with pressure backup	Flow with pressure backup
Bias/base flow range, L/min	None	0.4-5	10-20	10-20
Pressure slope/ramp adjustment	Auto/adjustable	0-9 relative scale	No	No
Sigh	Adjustable	Yes	Yes	Yes
100% O ₂	Not specified	Yes, adjustable % increase	Yes	Yes
Others	None specified	Advanced settings include flow cycle %, PSV flow %, PSVT maximum, volume limit, V sync, AAC	Advanced settings include PC flow cycle, PSV flow cycle, PSVT maximum	Advanced settings include volume limit, PC flow cycle, PSVT maximum

This is the first of four pages covering the above model(s). These specifications continue onto the next three pages.

Product Comparison Chart

MODEL	VERSAMED iVent 201	VIASYS HEALTHCARE AVEA	VIASYS HEALTHCARE Vela	VIASYS HEALTHCARE Vela +
OPERATING MODES				
Assist/control				
Volume breaths	Yes	Yes	Yes	Yes
Pressure breaths	Yes	Yes	Yes	Yes
SIMV				
Volume breaths	Yes	Yes	Yes	Yes
Pressure breaths	Yes	Yes	Yes	Yes
Pressure support	Yes	Yes	Yes	Yes
Spontaneous/CPAP				
Pressure support	Yes	Yes	Yes	Yes
Apnea-backup vent	Yes	Yes	Yes	Yes
Combination modes	Not specified	PRVC, machine volume	No	PRVC
Responsive valve	Not specified	Yes	Not specified	Not specified
Bilevel/APRV	Not specified	Yes	No	No
Others	Adaptive flow, adaptive time, easy exhalation, NPPV	APRV/BiPhasic, TCPL, PRVC, Vsync, heliox	None specified	Noninvasive A/C, SIMV, and CPAP/PSV; PRVC modes, Vsync
MONITORED PARAMETERS				
Pressure				
PIP	Yes	Yes	Yes	Yes
MAP	Yes	Yes	Yes	Yes
PEEP	Yes	Yes	Yes	Yes
Volume				
Tidal	Yes	Yes	Yes	Yes
Minute	Yes	Yes	Yes (VTI, VTE)	Yes (VTI, VTE)
Spontaneous minute	Yes	Yes	Yes	Yes
FiO2	Yes	Yes	Yes	Yes
Respiratory rate	Yes	Yes	Yes	Yes
Inspiratory time	Yes	Yes	Yes	Yes
Expiratory time	Yes	Yes	Yes	Yes
IE	Yes	Yes	Yes	Yes
Others	Displayed leak % and alarm	MIP/P100, P-Flex maneuver, AutoPEEP, Bicare esophageal parameters, mL/kg, C stat, Cdyn, all parameters trended 24 hr	None specified	None specified
PATIENT ALARMS				
FiO2	Adjustable	Yes	Yes	Yes
Low minute volume	Adjustable	Yes	Yes	Yes
Low inspiratory pressure	Adjustable	Yes	Yes	Yes
High pressure	Adjustable	Yes	Yes	Yes
Loss of PEEP	Adjustable	Yes	Low pressure	Low pressure
Apnea	Adjustable	Yes	Yes	Yes
High continuous pressure/occlusion	Adjustable	Yes	Circuit fault	Circuit fault
Inverse IE	Yes	Yes	No	No
High respiratory rate	Adjustable	Yes	Yes	Yes
High minute volume	Adjustable	Yes	Yes	Yes
High PEEP	Yes	No	No	No
Breathing circuit disconnect	Not specified	Yes	Circuit fault	Circuit fault
Others	None specified	Low Vt, high Vt, high breath rate	High breath rate	High breath rate

This is the second of four pages covering the above model(s). These specifications continue onto the next two pages.

Product Comparison Chart

MODEL	VERSAMED iVent 201	VIASYS HEALTHCARE AVEA	VIASYS HEALTHCARE Vela	VIASYS HEALTHCARE Vela +
EQUIPMENT ALARMS				
Gas supply failure	Yes	Yes	Yes	Yes
Power failure	Yes	Yes	Yes	Yes
Vent inoperative	Yes	Yes	Yes	Yes
Low battery	Yes	Yes	40% and 20% warnings	40% and 20% warnings
Self-diagnostics	Yes	Yes	Yes	Yes
Others	Check sensor, leak	Fan failure, safety valve open, invalid gas ID, ILV disconnect, heliox loss	Test for lamp, leak, switch, alarm, and filter; touchscreen calibration	Test for lamp, leak, switch, alarm, and filter; touchscreen calibration
INTERFACING				
Output ports	RS232, RJ45, VGA	RS232, SVGA, parallel	RS232	RS232
Remote alarm/display	Yes	Yes	Nurse call, others	Nurse call, others
Analog output	Not specified	Yes	Direct video	Direct video
Report generation				
Display	Yes	Full color SVGA LCD	Full color SVGA LCD	Full color SVGA LCD
Hard copy	Yes	Printer output	HP 940 driver output	HP 940 driver output
Archival disk	Yes	Internal storage	Not specified	Not specified
Network	Not specified	RS232	BMI optional	BMI optional
DISPLAY TYPES	Integrated color LCD	Graphics SVGA color LCD	Graphics SVGA color LCD	Graphics SVGA color LCD
DATA DISPLAYED	Numbers, waveforms, loops, trends, event log, alarms, history	Waveform and loops on monitor with trends	User configurable	User configurable
PNEUMATIC POWER				
Compressor	Turbine	Battery-backed scroll pump	Battery-backed turbine	Battery-backed turbine
Compressed gases	O2	O2, air, heliox	O2	O2
Pressure ranges	40-60 psig	20-80 psi	40-80 psig	40-80 psig
Turbine/piston	Not specified	Scroll pump	Turbine	Turbine
LINE POWER, VAC	100-240, 50/60 Hz	100/120/230/240	90-240, 47-65 Hz	90-240, 47-65 Hz
Current, amps	2.0 (maximum)	3 (120 VAC)	2	2
Watts	200	360	36-220	36-220
INTERNAL BATTERY				
Type (number)	Yes	Yes	Yes	Yes
	Sealed lead-acid 12 V (not specified)	Ni-MH	Ni-MH (not specified)	Ni-MH (not specified)
Operating time, hr	Up to 2	2	6	6
Rechargeable	Yes	Yes	Yes	Yes
Recharging time, hr	Not specified	4	8	8
EXTERNAL BATTERY				
Type (number)	Yes	Yes	None	None
	Not specified	Sealed lead-acid	NA	NA
Operating time, hr	6-8	6, plus 2 internal; 8 in total	NA	NA
Rechargeable	Not specified	Yes	NA	NA
Recharging time, hr	Not specified	12	NA	NA
H x W x D, cm (in)	33 x 24 x 26 (13 x 9.4 x 10.2)	43.2 x 40.6 x 26.7 (17 x 16 x 10.5)	30.5 x 33 x 36.8 (12 x 13 x 14.5)	30.5 x 33 x 36.8 (12 x 13 x 14.5)
WEIGHT, kg (lb)	10 (22.1)	≤36.3 (80) with compressor	17.2 (38)	17.2 (38)

This is the third of four pages covering the above model(s). These specifications continue onto the next page.

Product Comparison Chart

MODEL	VERSAMED iVent 201	VIASYS HEALTHCARE AVEA	VIASYS HEALTHCARE Vela	VIASYS HEALTHCARE Vela +
PURCHASE INFORMATION				
List price	Not specified	\$22,995-32,995 depending on options	\$13,995-17,995	\$13,995-17,995
Warranty	1 year	2 years	2 years; 5 years or 40,000 hr on turbine	2 years; 5 years or 40,000 hr on turbine
Service				
Factory/on-site	Yes/yes	Yes	Yes/yes	Yes/yes
Training	Yes	Class at factory	Class at factory	Class at factory
3rd-party service	Yes	In some areas	In some areas	In some areas
Parts availability	Yes	Yes	Yes	Yes
Delivery time, ARO	Not specified	30-45 days	30-45 days	30-45 days
Year first sold	Not specified	2002	2002	2002
Number sold				
USA/worldwide	Not specified	>2,000/>1,500	Not specified/>6,500	Not specified/>6,500
Fiscal year	January to December	January to December	January to December	January to December
OTHER SPECIFICATIONS	MRI compatible; rotational control knob interface; VGA output for secondary screen option; integral O2 sensor; battery backup; software upgrades via PC. Meets requirements of IEC 60601, CSA, and UL.	Tank holder; esophageal balloon/tracheal catheter monitoring for adult/pediatrics on comprehensive units; proximal hotwire flow sensing on deluxe units; proximal variable orifice flow sensing on comprehensive units; optional compressor and heliox on standard model; optional external battery on deluxe stand.	Low pressure (flowmeter); O2 inlet connection; synchronous nebulizer drive; touchscreen controls; transport cart option with cylinder holders.	Low pressure (flowmeter); O2 inlet connection; synchronous nebulizer drive; touchscreen controls; optional transport cart with cylinder holders.
LAST UPDATED	April 2006	April 2006	April 2006	April 2006
Supplier Footnotes				
Model Footnotes				
Data Footnotes				

Product Comparison Chart

MODEL	VIASYS HEALTHCARE Vela Comprehensive
WHERE MARKETED	Worldwide
FDA CLEARANCE	Yes
CE MARK (MDD)	Yes
PATIENT TYPE	Adult, pediatric
CONTROLS	
Tidal volume, mL	50-2,000
Inspiratory flow, L/min	10-140/180 spontaneous maximum
Inspiratory pressure, cm H ₂ O	0-100
Respiratory rate, breaths/min	2-80
Inspiratory time, sec	0.3-10
Expiratory time, sec	Depends on rate
IE ratio	4:1 maximum inverse
Inspiratory hold/plateau	Off, 0.1-2 sec
Expiratory hold	Manual
FiO ₂ , %	21-100
Manual breath	Yes
PEEP/CPAP, cm H ₂ O	0-35
Pressure support, cm H ₂ O	Off, 1-60
Nebulizer	Off, 1 to 60 minutes
Trigger mechanism	Flow with pressure backup
Bias/base flow range, L/min	10-20
Pressure slope/ramp adjustment	No
Sigh	Yes
100% O ₂	Yes
Others	Advanced settings include volume limit, assured volume, PC flow cycle, PSV flow cycle, PSVT maximum, V sync

This is the first of four pages covering the above model(s). These specifications continue onto the next three pages.

Product Comparison Chart

MODEL	VIASYS HEALTHCARE Vela Comprehensive
OPERATING MODES	
Assist/control	
Volume breaths	Yes
Pressure breaths	Yes
SIMV	
Volume breaths	Yes
Pressure breaths	Yes
Pressure support	Yes
Spontaneous/CPAP	
Pressure support	Yes
Apnea-backup vent	Yes
Combination modes	PRVC, assured volume
Responsive valve	Not specified
Bilevel/APRV	Yes
Others	APRV BiPhasic with PSV and time sync with volume limit, loops/trends, MIP/NIF, Vsync
MONITORED PARAMETERS	
Pressure	
PIP	Yes
MAP	Yes
PEEP	Yes
Volume	
Tidal	Yes
Minute	Yes (VTI, VTE)
Spontaneous minute	Yes
FiO2	Yes
Respiratory rate	Yes
Inspiratory time	Yes
Expiratory time	Yes
IE	Yes
Others	All parameters trended for 24 hr, flow/volume, pressure/volume, loops, spontaneous rate, spontaneous Ve
PATIENT ALARMS	
FiO2	Yes
Low minute volume	Yes
Low inspiratory pressure	Yes
High pressure	Yes
Loss of PEEP	Low pressure
Apnea	Yes
High continuous pressure/occlusion	Circuit fault
Inverse IE	No
High respiratory rate	Yes
High minute volume	Yes
High PEEP	No
Breathing circuit disconnect	Circuit fault
Others	High breath rate

This is the second of four
pages covering the above
model(s). These
specifications continue
onto the next two pages.

Product Comparison Chart

MODEL	VIASYS HEALTHCARE Vela Comprehensive
EQUIPMENT ALARMS	
Gas supply failure	Yes
Power failure	Yes
Vent inoperative	Yes
Low battery	40% and 20% warnings
Self-diagnostics	Yes
Others	Test for lamp, leak, switch, alarm, and filter; touchscreen calibration
INTERFACING	
Output ports	RS232
Remote alarm/display	Nurse call, others
Analog output	Direct video
Report generation	
Display	Full color SVGA LCD
Hard copy	HP 940 driver output
Archival disk	Not specified
Network	BMI optional
DISPLAY TYPES	Graphics SVGA color LCD
DATA DISPLAYED	User configurable
PNEUMATIC POWER	
Compressor	Battery-backed turbine
Compressed gases	O ₂
Pressure ranges	40-80 psig
Turbine/piston	Turbine
LINE POWER, VAC	90-240, 47-65 Hz
Current, amps	2
Watts	36-220
INTERNAL BATTERY	
Type (number)	Yes
Operating time, hr	Ni-MH (not specified)
Rechargeable	6
Recharging time, hr	Yes
	8
EXTERNAL BATTERY	
Type (number)	None
Operating time, hr	NA
Rechargeable	NA
Recharging time, hr	NA
H x W x D, cm (in)	30.5 x 33 x 36.8 (12 x 13 x 14.5)
WEIGHT, kg (lb)	17.2 (38)

This is the third of four pages covering the above model(s). These specifications continue onto the next page.

Product Comparison Chart

MODEL	VIASYS HEALTHCARE
	Vela Comprehensive
PURCHASE INFORMATION	
List price	\$13,995-17,995
Warranty	2 years; 5 years or 40,000 hr on turbine
Service	
Factory/on-site	Yes/yes
Training	Class at factory
3rd-party service	In some areas
Parts availability	Yes
Delivery time, ARO	30-45 days
Year first sold	2002
Number sold	
USA/worldwide	Not specified/>6,500
Fiscal year	January to December
OTHER SPECIFICATIONS	Low pressure (flowmeter); O2 inlet connection; synchronous nebulizer drive; touchscreen controls; optional transport cart with cylinder holders.
LAST UPDATED	April 2006
Supplier Footnotes	
Model Footnotes	
Data Footnotes	